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Simplified method for wetting on curved boundaries in conservative phase-field lattice-Boltzmann simulation of two-phase flows with large density ratios

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Abstract

A simplified method is proposed to implement wetting boundary condition on curved surfaces within the conservative phase-field lattice-Boltzmann (LB) simulation framework. It combines the idea of Huang et al. (Eur. Phys. J. E 41:17, 2018) to find the order parameter on the other side of the wall with the conservative Allen-Cahn equation (CACE) for interface evolution solved by the LB equations. It inherits the advantage of the original method using the Cahn-Hilliard equation to avoid complicated interpolations under different geometries. By using the CACE, the boundary condition for chemical potential is circumvented (making it more simplified), and the overshooting of the order parameter is also greatly suppressed, enabling it to simulate two-phase flows with solid objects of various shapes and wettabilities at large density and viscosity ratios. Several two-dimensional, axisymmetric and three-dimensional problems, including some previously studied by experiments, were simulated and the capability of the proposed method is demonstrated through its good agreement with theoretical predictions and/or experimental observations.

Keywords: *Lattice-Boltzmann Method, Two-Phase Flow, Conservative Phase Field, Wetting, Curved Boundary.*

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1 Introduction

Many natural phenomena and industrial problems involve two-phase incompressible viscous flows over curved walls, often with large density ratios (e.g., water droplet in air) [1]. One recent example that received much attention is the drop impact onto a sphere [2–8]. Significant efforts have been devoted to studying such complex two-phase flows in order to elicit the governing parameters and various underlying mechanisms by experiments and computer simulations [9]. Two-phase flow simulations require not only the solutions of equations for mass and momentum conservation, but also a proper equation for some indicator field distinguishing different fluids to describe the interface evolution. The numerical methods that can be used for the interface dynamics include the volume-of-fluid (VOF) method [10], level-set method (LSM) [11] and phase-field (diffuse-interface) method (PFM) [12, 13]. Among them, the PFM can handle topological changes naturally and has been applied in many multiphase flow simulations. Early PFM uses the Cahn-Hilliard equation (CHE) [14] for interface capturing [13]. The CHE contains fourth-order derivatives of the order parameter and has inherent diffusion that may lead to spontaneous drop shrinkage (to limited degree when the numerical parameters are properly set) [15]. The CHE-based PFM has been widely used for two-phase simulations with different discretization strategies (e.g., see [16–19]), among which some employed the lattice-Boltzmann model (LBM) to solve the CHE and the incompressible Navier-Stokes equations (NSEs) (e.g., [19]). By assuming a *hyperbolic tangent profile* for the order parameter across the interface and making use of it, [20] derived a phase-field equation for interface tracking. Based on [20] and inspired by the Allen-Cahn equation (ACE) [21], the conservative PFM was developed with excellent mass conservation property [22]. The conservative ACE (CACE) proposed in [22] resembles the governing equation in the conservative LSM [23]. Numerically, the CACE may be solved by different schemes. For instance, in [22] a dispersion-relation-preserving compact upwind scheme was employed, and in [24] a simple finite-difference scheme was adopted. The first LBM for the CACE was devised in [25] and was successfully applied for interface capturing. Since then, the CACE-based LBM has gained growing popularity in simulations of two-phase flows (e.g., see [26–34]) because its low dispersion property allows to handle large-density-ratio problems with relative ease (note the CHE-based LBM can also simulate large-density-ratio problems but requires more complicated schemes, e.g., see [35, 36]). Most of previous multiphase LBM works using either the CHE or CACE studied problems in regular domains, and just a few of them (e.g. [27, 32, 37–39] for binary fluids and [40, 41] for ternary fluids) investigated problems with wetting on curved surfaces. The limitations of some of the above works in handling curved solid boundaries were discussed in our previous studies (see [38, 42]). Here we only focus on recent publications not discussed before, including [32, 39]. In [32] the wetting boundary condition (WBC) adopts a surface energy form and essentially enforces a condition for the normal gradient of the order parameter on the curved wall, which requires complex interpolations even in two dimensions (2D). Note that the surface energy WBC assumes that the surface energy on the wall is a cubic function of the order parameter

(e.g., see [18, 34]). In [39] the staircase representation was used for the curved wall, and its accuracy is relatively low. There is another type of geometric WBC [43, 44] for PF-based simulations. [45] extended one of them (the characteristic interpolation in [44]) to curved surfaces but the interpolation is also relatively complex as it depends on the local configuration of the interface. The geometric WBC was derived purely based on geometric relations near the contact line [43, 44] and has also been introduced to the CACE-based LBM simulations [34]. In the same spirit of using the *hyperbolic tangent phase-field profile* as [20], we proposed in an earlier work [38] an alternative method for the WBC on curved walls that does not require complicated interpolations even in three dimensions (3D). And this method has been extended for simulation of ternary fluids in [42] recently. As we were not aware of the CACE then, we still used the CHE for interface evolution in [38, 42]. The purpose of the present work is to develop a CACE-based LBM simulation framework for large-density-ratio two-phase flows with immersed curved walls by incorporating the idea in [38]. Because the CACE can maintain the *hyperbolic tangent profile* of the order parameter (actually better than the CHE), the basis of the method in [38] is well preserved in CACE-based simulations. In addition, the CACE only involves the second-order derivatives of the order parameter and does not contain the gradient or laplacian of the chemical potential, the method can actually become simpler in CACE-based simulations. The proposed WBC on surfaces of general shapes does not rely on any form of surface energy and may be categorized into the geometric type according to its design spirit. However, it differs from other geometric WBCs because it makes full use of the hyperbolic tangent profile of the order parameter along the local normal direction of the interface. It requires the solution of a small inverse problem, but only on a few grid points near the contact lines (as will be shown below).

The remainder of this paper is organized as follows. Section 2 introduces the CACE-based LBM for simulation of two-phase flows at large density and viscosity ratios with particular focus on the WBC for curved immersed walls. The formulations in both Cartesian and cylindrical coordinates will be presented. Section 3 presents several test cases simulated by the proposed method, some of which include side-by-side comparisons with experiments. Section 4 concludes this paper with some discussions on future work.

2 Numerical Method

The present method for two-phase flow simulation consists of two parts: (1) the LBM for the CACE used for interface dynamics; (2) the LBM for the incompressible NSEs to simulate the hydrodynamics. They are coupled together through the velocity, the density and viscosity, and the surface tension force.

2.1 LBM for the CACE

The convective CACE can be written as,

$$\partial_t c + \mathbf{u} \cdot \nabla c = \nabla \cdot [M(\nabla c - \lambda \mathbf{n})], \quad (1)$$

where c is the order parameter varying between c_l in one fluid (e.g., the air) and c_h in the other fluid (e.g., the water), $\mathbf{u}$ is the fluid velocity, M is the (constant) mobility, $\mathbf{n} = \nabla c / |\nabla c|$ is the unit normal vector of the interface, and λ is given by $4(c - c_h)(c - c_l) / [W(c_l - c_h)]$ with W being the interface thickness. Across a flat interface c assumes a hyperbolic tangent profile $c(\zeta) = 0.5[(c_h + c_l) + (c_h - c_l) \tanh(\frac{-\zeta + \zeta_{\text{itf}}}{W/2})]$ (under the conditions $c(\infty) = 0$ and $c(-\infty) = 1$) where ζ is the coordinate along the interface normal $\mathbf{n}$, and ζ_{itf} is the position where $c = c_{\text{itf}} = 0.5(c_l + c_h)$. The chemical potential μ is given by,

$$\mu = 4 \frac{12\sigma}{W(c_h - c_l)^4} (c - c_l)(c - c_h)(c - c_{\text{itf}}) - \frac{3}{2} \frac{\sigma W}{(c_h - c_l)^2} \nabla^2 c, \quad (2)$$

and the surface tension force can be expressed as $\mathbf{F}_{ST} = \mu \nabla c$. In the present work, we choose $c_l = 0$ and $c_h = 1$, and the interface is represented by the contour at $c = 0.5$. Since we consider incompressible flows with $\nabla \cdot \mathbf{u} = 0$, we can rewrite eq. (1) as $\partial_t c + \nabla \cdot (c\mathbf{u}) = \nabla \cdot [M(\nabla c - \lambda \mathbf{n})]$.

In Cartesian coordinates, the CACE can be solved by using the following LBEs with the single-relaxation-time (SRT) collision model [25, 46],

$$g_i(\mathbf{x} + \mathbf{e}_i \delta_t, t + \delta_t) - g_i(\mathbf{x}, t) = -\frac{1}{\tau_g} [g_j(\mathbf{x}, t) - g_j^{eq}(\mathbf{x}, t)], \quad (3)$$

where g_i and g_i^{eq} are the distribution functions (DFs) and equilibrium DFs along the direction of the lattice velocity $\mathbf{e}_i$ ($i = 0, 1, \dots, b$), w_i is the weight for the direction along $\mathbf{e}_i$, c_s is the lattice sound speed ($c_{LBE} = \sqrt{3}c_s$ for the D2Q9 and D3Q19 velocity models in 2D and 3D), δ_t is the time step. The equilibrium DF is given by,

$$g_i^{eq} = c\Gamma_i + w_i \frac{M}{c_s^2} \lambda e_{i\alpha} n_\alpha, \quad (4)$$

with Γ_i being the dimensionless DF given by,

$$\Gamma_i = w_i \left[1 + \frac{1}{c_s^2} e_{i\alpha} u_\alpha + \frac{1}{2c_s^4} e_{i\alpha} u_\alpha e_{i\beta} u_\beta - \frac{1}{2c_s^2} u_\alpha u_\alpha \right]. \quad (5)$$

The order parameter is obtained from the DFs as,

$$c = \sum_i g_i. \quad (6)$$

For axisymmetric (pseudo 2D) problems with no azimuthal flows, the CACE is rewritten as [31],

$$\partial_t(rc) + \partial_\alpha(rcu_\alpha + Mc\delta_{\alpha r}) = \partial_\alpha[M\partial_\alpha(rc) - Mr\lambda n_\alpha], \quad (7)$$

where r is the coordinate in the radial direction, and $\delta_{\alpha r} = 1$ when $\alpha = r$ and vanishes otherwise. The axisymmetric CACE is solved by the axisymmetric LBM employing the following LBEs using the multiple-relaxation-time (MRT) collision model [31],

$$g_i(\mathbf{x} + \mathbf{e}_i\delta_t, t + \delta_t) - g_i(\mathbf{x}, t) = -(\mathbf{M}^{-1}\mathbf{S}_g\mathbf{M})_{ij}[g_j(\mathbf{x}, t) - g_j^{eq}(\mathbf{x}, t)] + \delta_t G_i(\mathbf{x}, t), \quad (8)$$

where $\mathbf{M}$ is a matrix that transforms the vector of DFs into a vector of moments [47],

$$\mathbf{M} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -4 & -1 & -1 & -1 & -1 & 2 & 2 & 2 & 2 \\ 4 & -2 & -2 & -2 & -2 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & -2 & 0 & 2 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \\ 0 & 0 & -2 & 0 & 2 & 1 & 1 & -1 & -1 \\ 0 & 1 & -1 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 \end{bmatrix}, \quad (9)$$

$\mathbf{M}^{-1}$ is its inverse that transforms the moments back into the DFs, $\mathbf{S}_g = \text{diag}(1, 1, 1, s_g, s_g, s_g, s_g, 1, 1)$ is the diagonal relaxation matrix with $s_g = 1/\tau_g$ and $\tau_g = 0.5 + M/(c_s^2\delta_t)$, and $G_i(\mathbf{x}, t)$ is a source term which can be expressed as,

$$G_i = [\mathbf{M}^{-1}(\mathbf{I} - 0.5\mathbf{S}_g)\mathbf{M}]_{ij}R_j, \quad (10)$$

with $\mathbf{I}$ being the unit matrix and $R_j = \frac{1}{c_s^2}w_j e_{j\alpha}[\partial_t(rcu_\alpha + Mc\delta_{\alpha r}) + c_s^2 r n_\alpha \lambda]$. The equilibrium moments for g_j^{eq} are given by $[rc, -2rc, rc, rcu_z, -rcu_z, rcu_r + Mc, -(rcu_r + Mc), 0, 0]^T$. In this axisymmetric LBM, the order parameter is obtained from

$$c = \frac{1}{r} \sum_i g_i. \quad (11)$$

85 In the above, it is seen that the 1st- and 2nd-order spatial derivatives of the order parameter are required.

For axisymmetric problems, they are evaluated based on the D2Q9 model as follows.

First, the computational nodes are categorized into two types,

- *boundary nodes* having at least one neighbor out of the eight being on the other side of the wall boundary;
- *bulk nodes* having all eight neighbors on the same side of the boundary.

For *boundary nodes*, the derivatives are obtained by the 2^{nd} -order isotropic scheme as,

$$\frac{\partial c}{\partial x_\alpha} = \frac{1}{c_s^2 \delta_t} \sum_{i=1}^8 w_i e_{i\alpha} c_v(\mathbf{x} + \mathbf{e}_i \delta_t), \quad (12)$$

$$\frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2} = \frac{6}{\delta_x^2} \left[\sum_{i=1}^8 w_i c_v(\mathbf{x} + \mathbf{e}_i \delta_t) - (1 - w_0) c(\mathbf{x}) \right]. \quad (13)$$

where $c_v(\mathbf{x} + \mathbf{e}_i \delta_t)$ is the order parameter at the node $\mathbf{x} + \mathbf{e}_i \delta_t$ set according to the following rules,

$$c_v(\mathbf{x} + \mathbf{e}_i \delta_t) = \begin{cases} c(\mathbf{x} + \mathbf{e}_i \delta_t) & \text{if } \mathbf{x} + \mathbf{e}_i \delta_t \text{ is on the same side as } \mathbf{x} \\ c_{vb}(\mathbf{x} + \mathbf{e}_i \delta_t) & \text{if } \mathbf{x} + \mathbf{e}_i \delta_t \text{ is on the other side of } \mathbf{x} \end{cases}, \quad (14)$$

with $c_{vb}(\mathbf{x} + \mathbf{e}_i \delta_t)$ being the order parameter set according to the BC on curved walls using the contact angle (to be described below). For *bulk nodes*, the spatial derivatives are obtained by the following high order isotropic scheme as [48],

$$\frac{\partial c}{\partial x_\alpha} = \frac{1}{6c_s^2 \delta_t} \sum_{i=1}^8 w_i e_{i\alpha} [8c(\mathbf{x} + \mathbf{e}_i \delta_t) - c_v(\mathbf{x} + 2\mathbf{e}_i \delta_t)], \quad (15)$$

$$\frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2} = \frac{1}{6c_s^2 \delta_t^2} \sum_{i=1}^8 w_i [16c(\mathbf{x} + \mathbf{e}_i \delta_t) - c_v(\mathbf{x} + 2\mathbf{e}_i \delta_t) - 15c(\mathbf{x})], \quad (16)$$

where $c_v(\mathbf{x} + 2\mathbf{e}_i \delta_t)$ is set in a way similar to $c_v(\mathbf{x} + \mathbf{e}_i \delta_t)$ in eq. (14). For a bulk node near the boundary (say, at $\mathbf{x}$), some of its neighbours (say, at $\mathbf{x} + \mathbf{e}_i \delta_t$) may be a boundary node and the node at $(\mathbf{x} + 2\mathbf{e}_i \delta_t)$ is on the other side of the wall. Then, $c_v(\mathbf{x} + 2\mathbf{e}_i \delta_t)$ is found from $c_{vb}((\mathbf{x} + \mathbf{e}_i \delta_t) + \mathbf{e}_i \delta_t)$ based on the boundary node at $\mathbf{x} + \mathbf{e}_i \delta_t$.

By setting $r = 1$ and removing the terms due to axisymmetric effects (e.g., terms containing Mc in R_j and the equilibrium moments) in the above axisymmetric LBM, one obtains a simplified MRT-LBM for the 2D CACE (the 2^{nd} -order isotropic scheme is used to calculate the spatial derivatives in 2D). In 3D, the spatial derivatives of c are evaluated using the 2^{nd} -order isotropic scheme based on the D3Q19 model (similar to eqs. (12) and (13)) for both bulk and boundary nodes. Once the gradient ∇c is obtained, the normal vector is calculated as $\mathbf{n} = \nabla c / (|\nabla c| + 10^{-12})$ where 10^{-12} is added to avoid division-by-zero. Near solid boundaries, half-way bounce-back is applied for g_i to avoid possible mass leakage [27].

Now we introduce the algorithm to find c_{vb} in eq. (14) (similar to that in [38]). To begin with, let us examine a 3D drop on a wall as shown in fig. 1a. The contact angle on the wall is θ_w . In the bulk region (with $c = 1$ inside the drop or $c = 0$ outside the drop), it is easy to determine c_{vb} . The wetting boundary condition is mainly used for points in the contact line region (i.e., points near the wall in the interfacial region with $0 < c < 1$). The local wall normal and tangential unit vectors are $\mathbf{n}_w$ and $\mathbf{t}_w$ (note that $|\frac{\partial c}{\partial \mathbf{t}_w}|$ is the maximum among all vectors perpendicular to $\mathbf{n}_w$). In the plane determined by $\mathbf{n}_w$ and $\mathbf{t}_w$, one can find the direction normal to the local interface $\mathbf{n}_{if}$, along which the order parameter obeys the hyperbolic tangent profile illustrated in fig. 1b. For ease of understanding, we describe the detailed steps in 2D as follows.

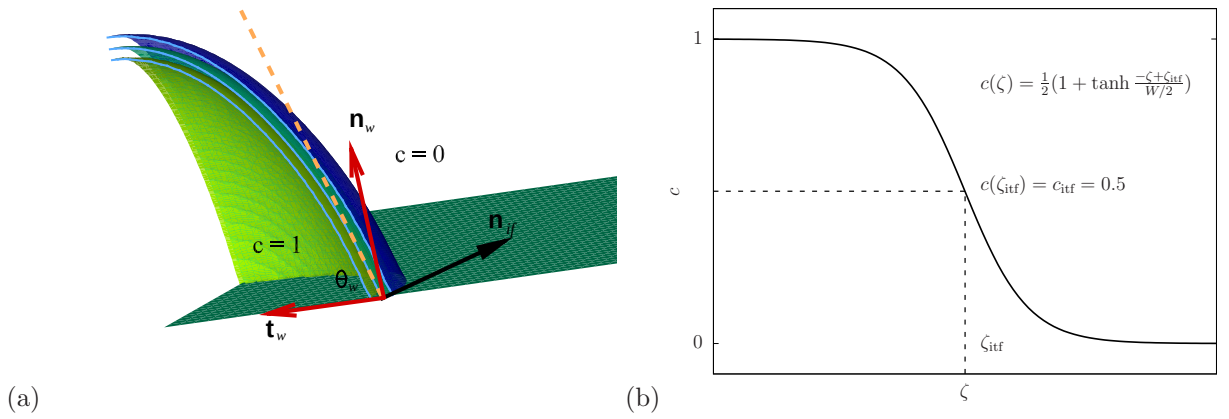


Figure 1: Illustration of (a) the contact line region where the wetting boundary condition is applied; (b) the hyperbolic tangent profile of the order parameter along the direction normal to the interface.

Figure 2 illustrates one situation in 2D for a boundary node at (x_0, y_0) . The eight neighbours connected with it through the vector $\mathbf{e}_i \delta_t$ ($i = 1, 2, \dots, 8$) are categorized into **Group-S** ("S" means on the *same* side of the wall) and **Group-O** ("O" means on the *other* side of the wall). In practice, whether a neighbour belongs to Group-S or Group-O is determined based on whether the link vector is cut by the wall (the nodes with the unknown values c_{vb} obviously belong to Group-O). Take the neighbour at (x_v, y_v) connected with (x_0, y_0) by the vector $\mathbf{e}_8 \delta_t$ as an example. We use the following steps to find $c_{vb}(x_v, y_v)$.

- First, as shown in fig. 2a, we calculate the local normal and tangential vectors of the wall from $\mathbf{n}_w = (x_0 - x_{0pw}, y_0 - y_{0pw})/s_0$ (the point (x_{0pw}, y_{0pw}) is the projection of (x_0, y_0) onto the wall, and s_0 is the distance between (x_{0pw}, y_{0pw}) and (x_0, y_0)) and $\mathbf{t}_w = \mathbf{n}_w e^{i(-\frac{\pi}{2})}$. With $\mathbf{t}_w$ found, we proceed to determine the sign of the tangential gradient of c by using $\frac{\partial c}{\partial \mathbf{t}_w} = \mathbf{t}_w \cdot \nabla c = t_{w,x} \frac{\partial c}{\partial x} + t_{w,y} \frac{\partial c}{\partial y}$ based on the values of c at selected points in Group-S (note the accuracy of $\frac{\partial c}{\partial x}$ and $\frac{\partial c}{\partial y}$ is not important because we only need to know the sign), and then determine the direction of the characteristic lines parallel to the interface (for instance, $\mathbf{t}_{if} = \mathbf{t}_w e^{i\theta_w}$ if $\frac{\partial c}{\partial \mathbf{t}_w} > 0$ as in fig. 2), and further to calculate the direction normal to the

$$\rho(c) \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mathbf{F}_{ST} + \nabla \cdot [\eta(c)(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)] \quad (18)$$

where p is the hydrodynamic pressure, $\rho(c)$ is the density linearly interpolated from the order parameter c as $\rho(c) = \frac{c-c_l}{c_h-c_l}\rho_L + \frac{c_h-c}{c_h-c_l}\rho_G$ (ρ_L and ρ_G are the densities of the liquid and gas), and $\eta(c) = \rho(c)\nu(c)$ is the dynamic viscosity with $\nu(c)$ being the kinematic viscosity found from that of the liquid ν_L , and of the gas ν_G , as $[\nu(c)]^{-1} = \frac{c-c_l}{c_h-c_l}\nu_L^{-1} + \frac{c_h-c}{c_h-c_l}\nu_G^{-1}$ [19]. The dynamic viscosities of the liquid and gas are $\eta_L = \rho_L\nu_L$ and $\eta_G = \rho_G\nu_G$, respectively.

In Cartesian coordinates, the LBEs using the SRT collision model for eqs. (17) and (18) read [49],

$$f_i(\mathbf{x} + \mathbf{e}_i\delta_t, t + \delta_t) - f_i(\mathbf{x}, t) = -\frac{1}{\tau_f}(f_i - f_i^{eq}) + \left(1 - \frac{1}{2\tau_f}\right)(\mathbf{e}_i - \mathbf{u}) \cdot [\nabla \rho c_s^2(\Gamma_i - \Gamma_i(0)) + \mathbf{F}_{ST}\Gamma_i]\delta_t, \quad (19)$$

where f_i and f_i^{eq} are the DFs and equilibrium DFs for the hydrodynamic variables with f_i^{eq} given by,

$$f_i^{eq} = w_i \left[p + \rho c_s^2 \left(\frac{1}{c_s^2} e_{i\alpha} u_\alpha + \frac{1}{2c_s^4} (e_{i\alpha} e_{i\beta} - c_s^2 \delta_{\alpha\beta}) u_\alpha u_\beta \right) \right], \quad (20)$$

and the relaxation parameter τ_f is related to the kinematic viscosity as $\nu = c_s^2(\tau_f - 0.5)\delta_t$. The macroscopic variables are calculated as,

$$p = \sum_i f_i + \frac{1}{2}\delta_t(\mathbf{u} \cdot \nabla \rho c_s^2), \quad (21)$$

$$\rho \mathbf{u} = \frac{1}{c_s^2} \sum_i f_i \mathbf{e}_i + \frac{1}{2}\delta_t \mathbf{F}_{ST}. \quad (22)$$

The axisymmetric LBM employed here is modified from that in [50]. For axisymmetric problems without azimuthal flows, the target NSEs to be recovered by the axisymmetric LBEs are,

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_\alpha}(\rho u_\alpha) = -\frac{\rho u_r}{r}, \quad (23)$$

$$\rho \left(\frac{\partial u_\alpha}{\partial t} + u_\beta \frac{\partial u_\alpha}{\partial x_\beta} \right) = -\frac{\partial p}{\partial x_\alpha} + F_{ST,\alpha} + \frac{\partial}{\partial x_\beta} \left[\eta \left(\frac{\partial u_\alpha}{\partial x_\beta} + \frac{\partial u_\beta}{\partial x_\alpha} \right) \right] + F_{\text{axisym},\alpha}, \quad (24)$$

where $F_{\text{axisym},\alpha}$ is given by,

$$F_{\text{axisym},\alpha} = \frac{\eta}{r} \left(\frac{\partial u_\alpha}{\partial r} + \frac{\partial u_r}{\partial x_\alpha} \right). \quad (25)$$

Note that eq. (25) contains one less term as compared with that in [50] because we use a different form of surface tension force and the additional term therein is taken into account through the chemical potential μ . For axisymmetric problems μ is still given by eq. (2) but with the Laplacian changed as,

$$\nabla_{\text{axisym}}^2 c = \frac{\partial^2 c}{\partial z^2} + \frac{\partial^2 c}{\partial r^2} + \frac{1}{r} \frac{\partial c}{\partial r}. \quad (26)$$

The axisymmetric LBEs for eqs. (23) and (24) read [50],

$$f_i(\mathbf{x} + \mathbf{e}_i \delta_t, t + \delta_t) - f_i(\mathbf{x}, t) = -\frac{1}{\tau_f}(f_i - f_i^{eq}) + \left(1 - \frac{1}{2\tau_f}\right)[S_i + S'_i + S''_i]\delta_t \quad (27)$$

where $S_i = (\mathbf{e}_i - \mathbf{u}) \cdot [\nabla \rho c_s^2 (\Gamma_i - \Gamma_i(0)) + \mathbf{F}_{ST} \Gamma_i]$ is a source term having the same form as in eq. (19), S'_i and S''_i are two additional source terms to account for the axisymmetric effects [50],

$$S'_i = w_i c_s^2 \left(-\frac{\rho u_r}{r} \right), \quad S''_i = (\mathbf{e}_i - \mathbf{u}) \cdot \left(\mathbf{F}_{\text{axisym}} - \frac{\rho u_r \mathbf{u}}{r} \right) \Gamma_i. \quad (28)$$

As compared with [50], the hydrodynamic pressure gradient ∇p (multiplied by the term $\Gamma_i - \Gamma_i(0)$) is neglected here (see [19] for details). In the axisymmetric LBM, the pressure and velocity are calculated as,

$$p = \sum_i f_i + \frac{1}{2} \delta_t \left(\mathbf{u} \cdot \nabla \rho c_s^2 - \rho c_s^2 \frac{u_r}{r} \right), \quad (29)$$

$$\rho \mathbf{u} = \frac{1}{c_s^2} \sum_i f_i \mathbf{e}_i + \frac{1}{2} \delta_t \left(\mathbf{F}_{ST} + \mathbf{F}_{\text{axisym}} - \frac{\rho u_r \mathbf{u}}{r} \right). \quad (30)$$

Because eqs. (29) and (30) are coupled, the pressure and velocity at the new time are obtained iteratively. The force due to axisymmetric effects $\mathbf{F}_{\text{axisym}}$ involves some velocity gradient terms and they are calculated by the 2^{nd} -order central difference schemes.

To improve stability, the MRT collision model [47] was used for axisymmetric and 2D problems and the weighted MRT (WMRT) model [29] was used for the 3D LBM. The MRT model in the axisymmetric / 2D LBM for NSEs is similar to that for the axisymmetric / 2D CACE except that the diagonal relaxation matrix is changed to $\mathbf{S}_f = \text{diag}(1, 1, 1, 1, 1, 1, 1, s_f, s_f)$ with $s_f = 1/\tau_f$. The WMRT model in the LBM for the 3D NSEs follows that in [29] with the relaxation rates in the diagonal relaxation matrix set to 1 except those for the second-order moments which are set to s_f . On an immersed solid boundary of any shape, the interpolated bounce-back method is applied for f_i according to [51].

3 Results and Discussions

In this section, we present the numerical results of several two-phase incompressible viscous flow problems with solid walls immersed inside the domain at large density and viscosity ratios by using the above 2D, axisymmetric and 3D conservative PF LBM. For most cases, the reference length L_r is chosen as the drop radius R . Once a reference velocity U_r is selected, a reference time $T_r = L_r/U_r$ can be derived. From the surface tension and liquid viscosity one can derive a capillary-viscous velocity scale as $U_{cv} = \sigma/\eta_L$ and the corresponding capillary-

viscous time is $T_{cv} = L_r/U_{cv} = \eta_L R/\sigma$. We may also use the capillary-inertial velocity $U_{ci} = \sqrt{\sigma/(\rho_L R)}$ and the corresponding capillary-inertial time is $T_{ci} = R/U_{ci} = \sqrt{\rho_L R^3/\sigma}$. These two velocity and time scales are related to each other as $U_{ci} = Oh U_{cv}$ and $T_{ci} = T_{cv}/Oh$ where Oh is the Ohnesorge number $Oh = \eta_L/\sqrt{\rho_L R\sigma}$. The Reynolds number based on U_{cv} is $Re_{cv} = \rho_L U_{cv} L_r/\eta_L = \rho_L \sigma R/\eta_L^2 = 1/Oh^2$ (note Re_{cv} is also known as the Laplace number La). In conservative phase-field simulations, there are two numerical parameters, namely, the Cahn number $Cn = W/L_r$ and the Peclet number $Pe = U_c W/M$ where U_c is some characteristic velocity varying with the problem. Unless specified otherwise, we set the interface width as $W \approx 5\delta_x$ (commonly used in the literature, e.g., see [28, 30, 33]). This can provide sufficient resolution of the order parameter profile across the interface even for large-density-ratio two-phase flows. The Cahn number Cn is about 0.1 and the Peclet number Pe varies from $O(1)$ to $O(10)$ for different problems. For axisymmetric cases, symmetric BCs are applied for the lower boundary (the axis).

3.1 Drop spreading on an inclined wall

We first consider a special 2D boundary, i.e., a plate embedded in the fluid domain. The initial setup is illustrated in fig. 3. In the $x - y$ plane, the plate is represented by a slant line segment from $(x_{s,l}, y_{s,l}) = (0.5, 0.5)$ to $(x_{e,l}, y_{e,l}) = (7.5, 2)$. The angle formed by the x -axis and the plate is $\alpha \approx 12.1^\circ$ ($\tan \alpha = 3/14$). The spreading of a drop on this plate is simulated by the 2-D LBM. The contact angle on this plate is $\theta_w = 60^\circ$. The density and viscosity ratios are $r_\rho = 1000$ and $r_\nu = 0.04$ ($r_\eta = 40$). The initial drop center is at $(x_{cd}, y_{cd}) = (3.791, 2.228)$, and the contact point between the drop and the plate is at $(x_0, y_0) = (4, 1.25)$. The domain size is 8×4.5 and periodic boundary conditions are applied at all four domain boundaries. To assess the performance of the proposed WBC, two control simulations were also performed with the plate coinciding with the left domain boundary (in the $x' - y'$ plane of fig. 3) under the same parameters (except the domain). In the control simulations, the domain is 4×4 with symmetric boundary conditions applied at $y' = 0$ (i.e., only the upper half domain was used) and stationary wall boundary conditions applied at the other three boundaries. The geometric WBC was applied on the left boundary in one of the control simulations whereas the cubic surface energy (CubSE) WBC was applied in the other. Some shared parameters are $Re = 100$, $N_L = 37.5$, $N_t = 2250$, and $Cn = 0.1333$ (the interface thickness $W/\delta_x \approx 5$). Figure 4 compares several snapshots of the drop interface by the present simulation and by using the other two WBCs. It is seen from fig. 4 that the differences between the present interface shapes and those using either the geometric or CubSE WBC are small at all times. More careful examination reveals that the present results match those by the CubSE WBC slightly better.

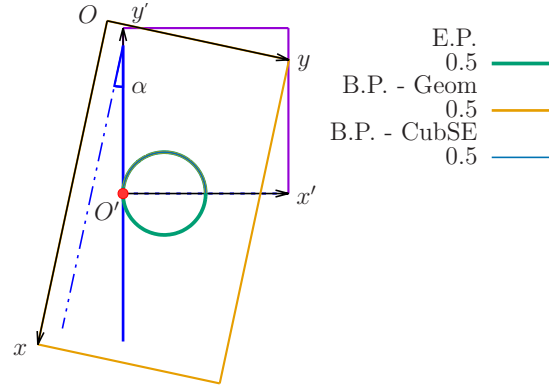


Figure 3: Initial setups for 2D drop spreading on a wall by two kinds of simulations overlaid so that the initial drop interfaces and wall positions overlap. In the $x - y$ plane, the drop spreads on an inclined plate embedded in the fluid domain with the initial contact point between the drop and the plate at (x_0, y_0) (the small red dot), and the WBC uses the present simplified method on curved walls. In the $x' - y'$ plane, the drop spreads on the left boundary and the WBC uses either the geometric (Geom) or cubic surface energy (CubSE) form. The two sets of coordinates are related through $x' = (x - x_0) \cos(\alpha + \frac{\pi}{2}) + (y - y_0) \sin(\alpha + \frac{\pi}{2})$ and $y' = -(x - x_0) \sin(\alpha + \frac{\pi}{2}) + (y - y_0) \cos(\alpha + \frac{\pi}{2})$. The solid blue line represents the inclined wall in the $x - y$ plane. The angle of inclination of the wall with respect to the x -axis is α . E.P. denotes "Embedded Plate" and "B.P." denotes boundary plate.

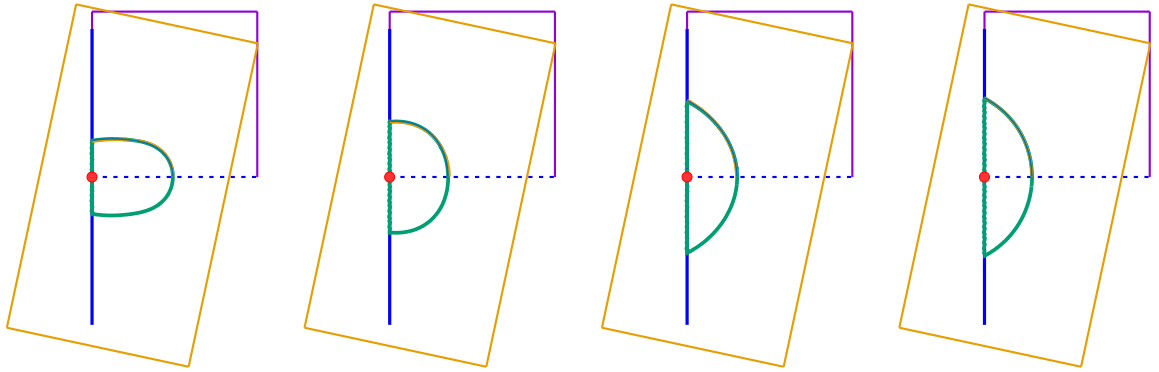


Figure 4: Snapshots of the interfaces at $t = 10, 20, 40$, and 60 (from left to right) by two kinds of simulations overlaid in the same manner as in fig. 3. Some shared parameters are $r_\rho = 1000$, $r_\nu = 0.04$ ($r_\eta = 40$), $Cn = 0.1333$, $N_L = 37.5$, $N_t = 2250$, $Re = 100$. The legends for different simulations are the same as in fig. 3 and omitted here for conciseness.

3.2 Drop spreading on a sphere

Next, we investigate the drop spreading on a sphere by using the axisymmetric LBM. Initially, a drop with a radius R is put close to a sphere with a radius R_s and contact angle θ_w . The initial centers of the drop and sphere are located at (x_{cd}, y_{cd}) and (x_{cs}, y_{cs}) , respectively. By setting $x_{cd} = x_{cs} + R_s + R$ and $y_{cd} = y_{cs} = 0$, we let the drop meet the sphere just at one point (in theory). The domain size is $L_x \times L_y$. Periodic BCs are used on the left and right boundaries and stationary wall is assumed for the upper boundary. Under the action of the surface tension force, the drop spreads on the sphere and eventually approaches an equilibrium state covering part of the sphere (see fig. 6 in [38] for an illustration of the equilibrium configuration). The height of the drop along the axis h_z was monitored during the spreading process. The Laplace number is fixed at $La = 100$ (the Ohnesorge number is $Oh = 0.1$).

We first consider a case at small density and viscosity ratios with $r_\rho = 10$ and $r_\eta = 10$ ($r_\nu = 1$). Some other parameters for this case are $R_s = 1$, $\theta_w = 60^\circ$, $(x_{cd}, y_{cd}) = (3.1, 0)$, $(x_{cs}, y_{cs}) = (1.1, 0)$, and $L_x \times L_y = 4.5 \times 2$. The simulation parameters are $N_L = 32$, $N_t = 192$ (to discretize T_{cv}), $Cn = 0.125$ ($W/\delta_x = 4$), $Pe_{U_{cv}} = 10$. This case was simulated in [38] by the 3D LBM using the CHE for interface evolution (solved by a finite-difference (FD) scheme) with the same spatiotemporal resolutions and Cn (W). Figure 5 compares the evolutions of the drop height h_z by the present axisymmetric LBM and by the 3D hybrid LB-FD method in [38]. One can find that the present axisymmetric results follow closely the 3D results.

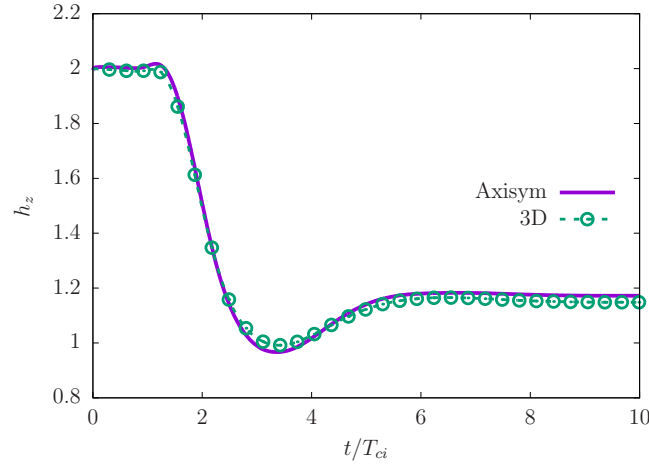


Figure 5: Evolutions of the drop height along the axis h_z by the axisymmetric LBM simulation and by the 3D hybrid LB-FD simulation in [38] (fig. 12 therein). The density and viscosity ratios are $r_\rho = 10$ and $r_\eta = 10$ ($r_\nu = 1$), and the Ohnesorge number is $Oh = 0.1$. The reference 3D results were obtained with the CHE for interface evolution in [38].

Besides, we study some cases at large density and viscosity ratios with $r_\rho = 1000$ and $r_\eta = 100$ ($r_\nu = 0.1$). The other parameters are $R_s = 0.85$, $(x_{cd}, y_{cd}) = (3.05, 0)$, $(x_{cs}, y_{cs}) = (1.2, 0)$, and $L_x \times L_y = 5 \times 2.5$. Four different contact angles were tested, including $\theta_w = 45^\circ$, 60° , 120° , and 135° . The simulation parameters are $N_L = 37.5$,

208 $N_t = 750$ (to discretize T_{cv}), $Cn = 0.1333$ ($W/\delta_x \approx 5$), $Pe_{U_{cv}} \approx 10$. Figure 6 compares the evolutions of the
 209 drop height for the four contact angles. Based on fig. 6, one can deduce that in all cases the drop experiences
 210 some oscillations during the initial spreading stage and then gradually settles down on the sphere. It seems
 211 that the drop approaches its equilibrium state faster on hydrophilic spheres than on hydrophobic spheres. This
 212 could be due to the larger contact area with the more hydrophilic sphere which helps to dissipate the kinetic
 213 energy of the drop faster. Figure 7 shows the interface shapes in equilibrium (at $t/T_{ci} = 20$ for $\theta_w = 45^\circ, 60^\circ$
 214 and $t/T_{ci} = 30$ for $\theta_w = 120^\circ, 135^\circ$) for the four contact angles of the sphere together with the theoretical
 215 predictions (see [38] for details on the analytical solution). It is seen that the present results are in reasonably
 216 good agreement with the theoretical ones for all cases.

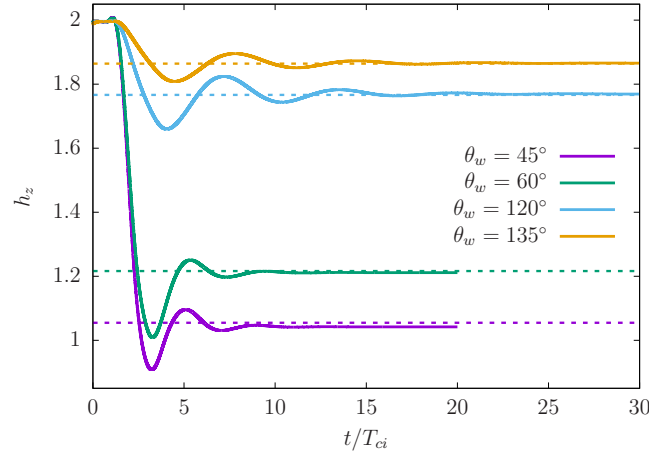


Figure 6: Evolutions of the drop height along the axis h_z by the axisymmetric LBM simulations. The density and viscosity ratios are $r_\rho = 1000$ and $r_\eta = 100$ ($r_\nu = 0.1$), and the Ohnesorge number is $Oh = 0.1$. The dashed lines are the corresponding theoretical predictions of the drop height in equilibrium for the four contact angles.

217 3.3 Drop spreading on an oblate/prolate spheroid

In addition to a sphere, we also studied drop spreading on an oblate/prolate spheroid under the axisymmetric condition. The problem setup, density and viscosity ratios are almost the same as those in the above (at large density ratios) except that the sphere is replaced by an oblate/prolate spheroid with the lengths of the semi-axes along the z - and r -axes being R_z and R_r . Specifically, we used $R_r = 1.25$ and $R_z = 0.75$ for an oblate spheroid and $R_r = 0.75$ and $R_z = 1.25$ for a prolate spheroid. Two contact angles $\theta_w = 60^\circ$ and 120° were considered for both types of spheroids. Other physical and numerical parameters are $La = 25$ ($Oh = 0.2$), $N_L = 37.5$, $N_t = 2250$ (to discretize T_{cv}), $Cn = 0.1333$ ($W/\delta_x \approx 5$), $Pe_{U_{cv}} \approx 10$. For this problem, we monitored the centroid velocity of the drop along the axis $V_{z,cm}$ which was calculated by,

$$V_{z,cm} = \frac{\int_{A|_{c>c_{itf}}} ru(r,z)drdz}{\int_{A|_{c>c_{itf}}} rdrdz} \approx \frac{\sum_{i,j|_{c_{i,j}>c_{itf}}} r_{i,j}u_{i,j}}{\sum_{i,j|_{c_{i,j}>c_{itf}}} r_{i,j}}, \quad (31)$$

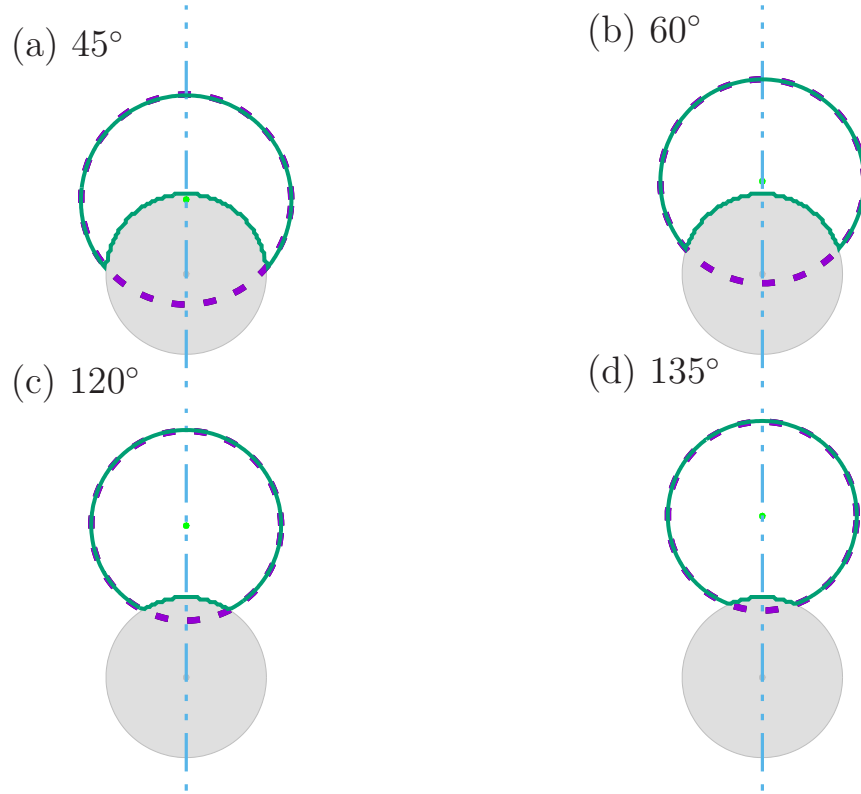


Figure 7: Equilibrium shapes of a drop on a sphere with different contact angles: (a) $\theta_w = 45^\circ$; (b) $\theta_w = 60^\circ$; (c) $\theta_w = 120^\circ$; (d) $\theta_w = 135^\circ$. The theoretical predictions are plotted in dashed lines and the present results are in solid lines. The plots are rotated by $\pi/2$ anticlockwisely and the right half (the lower half before rotation) is obtained by using the symmetric condition about the axis (the dash-dot-dot line). Such operations are also applied for the following plots in figs. 9, 10 and 12. The shaded part denotes the sphere.

where $A|_{c > c_{itf}}$ denotes the region with $c > c_{itf}$. Figure 8 compares the evolutions of the drop's centroid velocity in z -direction $V_{z,cm}$ for both spheroids with $\theta_w = 60^\circ$ and 120° . It is observed that the oscillation of drop velocity has the largest amplitude on the hydrophilic oblate spheroid; in contrast, it has the smallest amplitude on the hydrophobic prolate spheroid. In all four cases the drop velocity approaches zero after $t/T_{ci} \approx 20$. Figure 9 shows the interface shapes at $t/T_{ci} = 30$ (almost in equilibrium) for the four cases. The analytical solution for each case is also plotted in fig. 9 (the details on how to obtain the analytical solution are given in the Appendix A). It is seen from fig. 9 that the results obtained by the present axisymmetric LBM agree well with the analytical predictions for both the oblate and prolate spheroids under both hydrophobic and hydrophilic conditions.

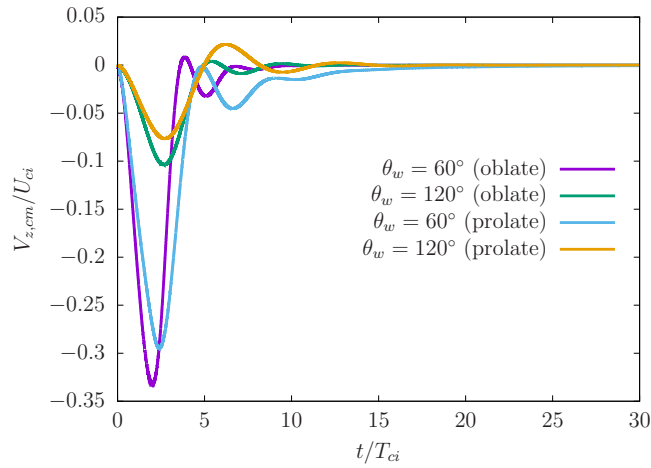


Figure 8: Evolutions of the centroid velocity of the drop along the axis $V_{z,cm}$ by the axisymmetric LBM simulations during the spreading on an oblate or prolate spheroid with $\theta_w = 60^\circ$ and 120° . The density and viscosity ratios are $r_\rho = 1000$ and $r_\eta = 100$ ($r_\nu = 0.1$), and the Ohnesorge number is $Oh = 0.2$.

3.4 Drop impact on a sphere

In this part, the drop impact on a sphere was simulated by the axisymmetric LBM. The initial setup is almost the same as that for drop spreading on a sphere except that the drop is given an initial velocity of magnitude U_{imp} towards the sphere. Usually, the impact Reynolds and Weber numbers are defined based on U_{imp} and the drop diameter $D = 2R$ as [52],

$$Re_{imp} = \frac{\rho U_{imp} D}{\eta_L}, \quad We_{imp} = \frac{\rho U_{imp}^2 D}{\sigma}. \quad (32)$$

The Ohnesorge number based on D is $Oh_D = \eta_L / \sqrt{\rho_L D \sigma} = \sqrt{We_{imp}} / Re_{imp} = Oh / \sqrt{2}$. For this problem, we introduce a characteristic time $T_{imp} = D / U_{imp}$. Stationary wall was assumed for the upper, left and right boundaries.

Two cases in [3] and [4] are considered here. In the first case, the sphere has a radius $R_s = 3.226$ and contact angle $\theta_w = 90^\circ$, the density and viscosity ratios are $r_\rho = 829$ and $r_\eta = 55.16$ ($r_\nu = 0.0665$) which are close to

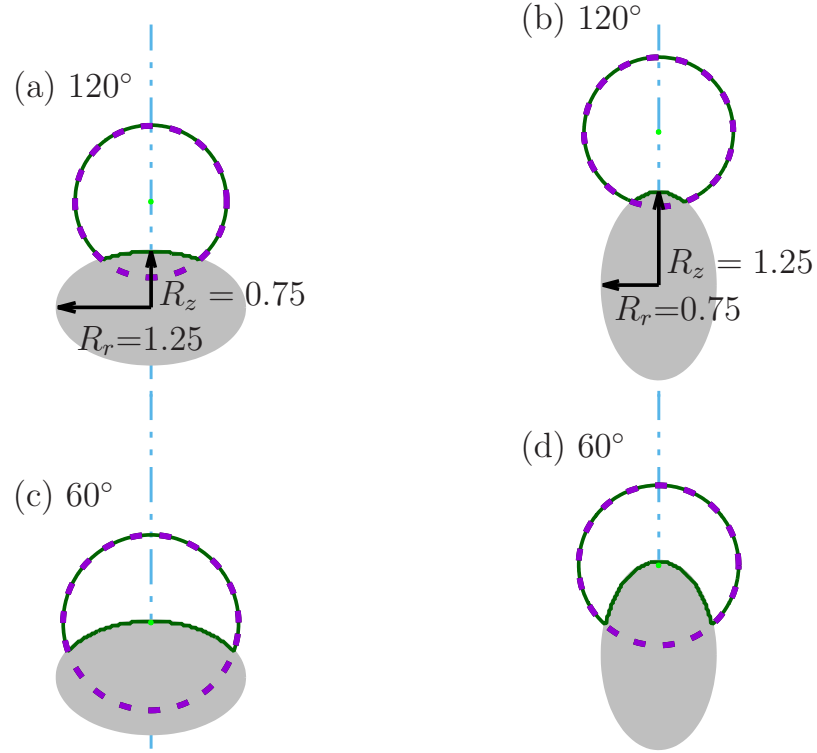


Figure 9: Equilibrium shapes of a drop on an oblate/prolate spheroid with different contact angles: (a) $\theta_w = 120^\circ$, $R_z = 0.75$, $R_r = 1.25$; (b) $\theta_w = 120^\circ$, $R_z = 1.25$, $R_r = 0.75$; (c) $\theta_w = 60^\circ$, $R_z = 0.75$, $R_r = 1.25$; (d) $\theta_w = 60^\circ$, $R_z = 1.25$, $R_r = 0.75$. The left column is for an oblate spheroid and the right is for a prolate spheroid. The analytical predictions are plotted in dashed lines and the present numerical results are in solid lines. The shaded part denotes the oblate/prolate spheroid.

the values in the experiments of [3] (fig. 4 therein). The impact Weber number is $We_{imp} = 8$ (same as [3]) and the Reynolds number is $Re_{imp} = 650$ (smaller than $Re_{imp} = 1349$ in [3]). The diameter-based Ohnesorge number is $Oh_D = 4.35 \times 10^{-3}$. Here we adjusted Re_{imp} to avoid instability that occurs at higher Re_{imp} (lower Oh_D). Similar practice was also done for the same purpose in a recent work [39], where a much lower value ($Re_{imp} = 113$) was used for the same problem. The domain size is $L_x \times L_y = 12 \times 5$. The numerical parameters are $N_L = 40$, $N_t = 20$ (to discretize T_{cv}), $Cn = 0.125$ ($W/\delta_x = 5$), $Pe_{U_{imp}} \approx 12.3$. Figure 10 shows five snapshots of the drop shape at the same (dimensionless) times as the experiments of [3]. Overall, the evolution of the drop morphology is similar to the experimental observations. But some differences can be also seen, especially in the recoil stage (at $t/T_{imp} = 3.094$ and 3.657). This is likely because the present simulation did not include the contact angle hysteresis. Here we only used one equilibrium contact angle (instead of the advancing and receding contact angles for different stages). For this case we also examined the spread factor D_{max}/D where D_{max} is the arc length on the sphere covered by the drop in the $r-z$ plane. Figure 11 shows the evolution of the spread factor by the present axisymmetric LBM simulation together with the experimental and numerical results from [3]. Note that the simulation in [3] used the 3D VOF model using dynamic contact angle. It is seen that the present evolution of D_{max}/D in general follows the experimental measurement and the agreement appears to be slightly better than the 3D VOF simulation in [3].

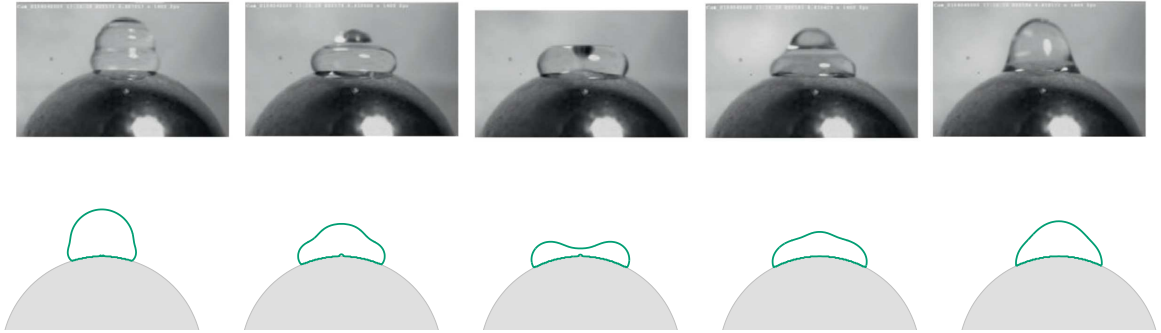


Figure 10: Snapshots of the drop shape during drop impact on a sphere at $t/T_{imp} = 0.563, 1.125, 1.969, 3.094$ and 3.657 (from left to right). The upper row is from [3] (Reproduced with permission from Chem. Eng. Sci. 100, 105 (2013). Copyright 2013 Elsevier Ltd.) and the lower row is by the present axisymmetric simulation.

In the second case, the sphere has a radius $R_s = 1.35$ and contact angle $\theta_w = 95^\circ$, the density and viscosity ratios are $r_\rho = 844.6$ and $r_\eta = 54.05$ ($r_\nu = 0.064$), and the impact Reynolds and Weber numbers are $Re_{imp} = 1638$ and $We_{imp} = 16.8$ ($Oh_D = 2.502 \times 10^{-3}$). These parameters almost match those in the experiments of [4] (fig. 1 therein). The domain size is $L_x \times L_y = 6 \times 4$. The numerical parameters are $N_L = 40$, $N_t = 20$ (to discretize T_{cv}), $Cn = 0.125$, $Pe_{U_{imp}} \approx 10.26$. Figure 12 presents four snapshots of the drop shape at the same moments as in the experiments [4]. One can find that the numerical predictions resemble the corresponding photos for this case. The agreement actually appears to be even better than the above case. The reason may be two-fold: (1) the impact physical parameters are not altered in the simulation; (2) the contact angle matches the advancing contact angle in the experiments and we only compared the spreading stage (as in fig. 1 of [4]).

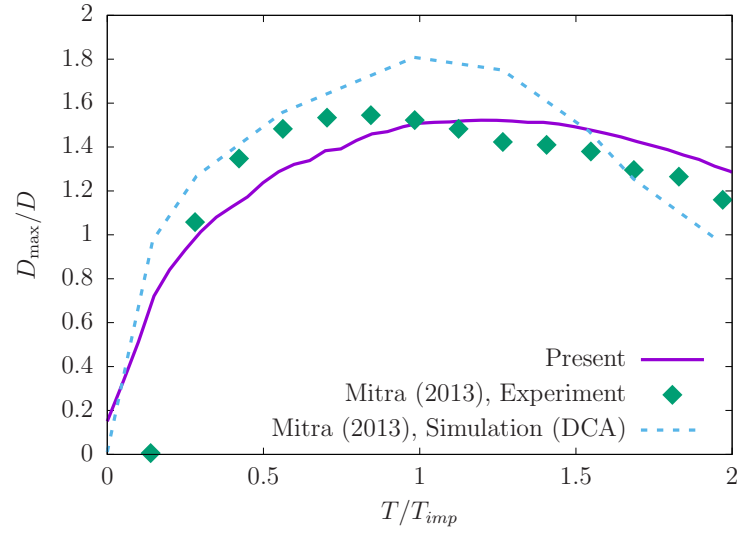


Figure 11: Evolution of the spread factor during drop impact on a sphere. The solid line is by the present axisymmetric LBM simulation, the symbols and dashed line are by experimental measurement and 3D VOF simulation using dynamic contact angle (DCA) in [3].

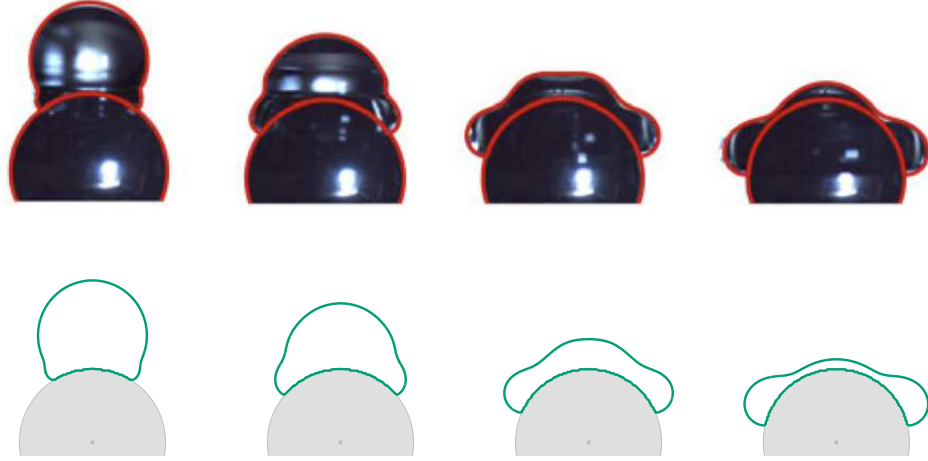


Figure 12: Snapshots of the drop shape during drop impact on a sphere at $t/T_{imp} = 0.4, 0.9, 1.8$ and 2.7 (from left to right). The upper row is from [4] (Reproduced with permission from J. Fluid Mech. 824, R3 (2017). Copyright 2017 Cambridge University Press.) and the lower row is by the present axisymmetric simulation.

3.5 Coalescence-induced droplet jumping from a round hydrophobic fiber

Lastly, we studied the coalescence-induced droplet jumping from a round hydrophobic fiber in 3D. The initial setup is shown in fig. 13 (panels (a) and (b) are for a symmetric case with two equal-sized droplets and panel (c) is for an asymmetric case). The fiber radius is r_f and the contact angle on the fiber surface is θ_f . The fiber axis is parallel to the z -axis and assumed to extend from $z = -\infty$ to $z = \infty$, and its projection in the $x - y$ plane is at (x_f, y_f) (for the symmetric case, $x_f = 0$). The problem is symmetric about the plane $z = 0$, and one can consider only the front half of the domain ($z > 0$). For two equal-sized droplets, the problem can be further simplified and only the quarter of the domain with $x > 0$ and $z > 0$ is used. The simulation domain is a box of size $L_x \times L_y \times L_z$. The initial radius of the droplet on the right (denoted as "droplet 1") is $R_1 = R$ and its initial center is fixed at $(x_f + R, y_d, 0)$ with $y_d = y_f + \sqrt{r_f^2 - 2r_f R \cos \theta_f}$ which satisfies the condition that the initial shape matches the equilibrium configuration corresponding to the contact angle θ_f in the $x - y$ plane at $z = 0$. The initial shape specified above deviates slightly from the true equilibrium shape of a droplet on a round fiber. This should be acceptable because the droplet radius is much larger than the fiber radius and the volume of the droplet cut by the fiber scales with $(r_f/R)^3$ in 3D, plus the fiber is hydrophobic with a relatively large contact angle. The initial radius of the left droplet ("droplet 2") is R_2 and its initial center is fixed at $(x_{d2}, y_{d2}, 0)$ with x_{d2} and y_{d2} determined based on two requirements (see fig. 13c): (1) it is tangent to the right droplet; (2) it also satisfies the above condition related to the contact angle. Symmetric boundary conditions are applied at $z = 0$, L_z . Periodic boundary conditions are applied at $y = 0$, L_y . For the symmetric case, symmetric boundary conditions are applied at $x = 0$, L_x , whereas for the asymmetric case periodic boundary conditions are applied at $x = 0$, L_x . The boundary conditions on the fiber surface are those for curved surfaces described above. Table 1 lists the properties of the liquid and gas, the reference quantities (reference density, length, velocity and time ρ_r , L_r , U_r , T_r) and dimensionless parameters (the density ratio r_ρ , dynamic viscosity ratio r_η , Ohnesorge number Oh , contact angle of the fiber surface θ_f , droplet-to-fiber radius ratio Γ , and the ratio of the right droplet radius over the left one r_{12}). These parameters match those for typical cases in the experiments of [53] except that for the asymmetric case the Ohnesorge number Oh is increased (the droplet radius is decreased) to reduce the computational time. It is noted that the increased Oh in our simulation is still low enough to ensure that the problem is in the capillary-inertial regime (as in the simulation in [53]). Previously, the symmetric case was also simulated by a hybrid level-set immersed-boundary curvature-boundary method in [54] and recently by the moment of fluid method (a generalized VOF method) [55]. For this problem, all lengths, velocities and times are measured in L_r , U_r and T_r respectively. It is noted that here we use the right drop radius at the beginning as the reference length L_r and the initial right droplet radius is $R = 1$. The fiber radius varies with Γ as $r_f = 1/\Gamma$ and the left droplet radius varies with r_{12} as $R_2 = 1/r_{12}$. The domain size is $4 \times 6 \times 3$ for the symmetric case, and $9 \times 6 \times 3$ for the asymmetric case. The time simulated is $[0, 6T_{ci}]$ for

the symmetric case, and $[0, 12.5T_{ci}]$ for the asymmetric case. The numerical parameters are $N_L = 40$, $N_t = 200$ (to discretize T_{cv}), $Cn = 0.125$, $Pe_{U_{cv}} = 100$.

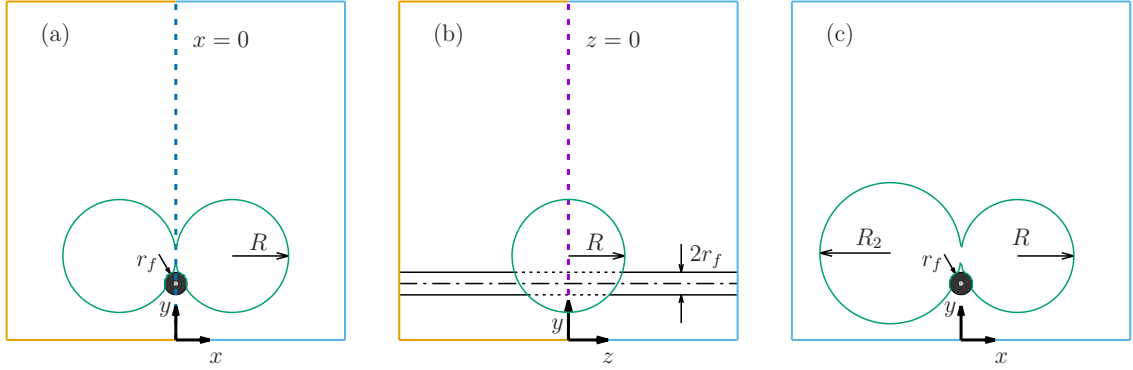


Figure 13: Initial setup for the 3-D coalescence-induced droplet jumping from a fiber: (a) front view (projection in the $x - y$ plane at $z = 0$) of the symmetric case; (b) left view (projection in the $y - z$ plane at $x = 0$) of the symmetric case; (c) front view of the asymmetric case (for illustration only). The initial center of the right droplet is fixed at $(x_f + R, y_d, 0)$. The initial center of the left droplet is at $(x_{d2}, y_{d2}, 0)$ such that it is tangent to the right one and also satisfies the contact angle condition in the $x - y$ plane. In (a) and (c), the filled black circle represents the fiber.

Table 1: Physical properties, reference quantities and dimensionless parameters for the study of coalescence-induced droplet jumping from a fiber. In the third column, the single values are shared by both the symmetric and asymmetric cases, and when two values are given, the left is for the symmetric case and the right is for the asymmetric case.

Quantity / Parameter	Choice / Definition	Value(s)
Liquid density (kg/m^3)	ρ_L	998
Gas density (kg/m^3)	ρ_G	1.184
Liquid dynamic viscosity ($mPa \cdot s$)	η_L	1.071
Gas dynamic viscosity ($mPa \cdot s$)	η_G	0.01849
Surface tension (mN/m)	σ	72.7
Initial (right) droplet radius (μm)	R	249, 100
Fiber radius (μm)	r_f	46.0, 15.35
Fiber contact angle	θ_f	120°
Reference density (kg/m^3)	$\rho_r = \rho_L$	998
Reference length (μm)	$L_r = R$	249, 100
Reference velocity (m/s)	$U_r = U_{ci} = \sqrt{\sigma/(\rho_L R)}$	0.5409, 0.8535
Reference time (ms)	$T_r = T_{ci} = R/U_{ci}$	0.4604, 0.1172
Density ratio	$r_\rho = \rho_L/\rho_G$	842.9
Viscosity ratio	$r_\eta = \eta_L/\eta_G$	57.92
Ohnesorge number	$Oh = \eta_L/\sqrt{\rho_L \sigma R}$	0.00797, 0.0126
Droplet-to-fiber radius ratio	$\Gamma = R/r_f$	5.413, 6.513
Droplet radius ratio r_{12}	R/R_2	1, 0.5

Figure 14 shows ten snapshots of the droplets for the symmetric cases by the present simulation and from the other two sources at the same times. It is seen that both numerical simulations predict the evolution of the droplet's shape very well before the droplet leaves the fiber. There exist some differences with regard to the time for the droplet to get off the fiber: both simulations give a slightly early departure time (though the two sets of simulation results appear to be closer). This may be related to the exclusion of contact angle hysteresis

effect in both simulations. In the experiment the fiber surface has an advancing contact angle of $\theta_A = 121^\circ$ and receding contact angle $\theta_R = 108^\circ$. The hysteresis effect delays the departure of the droplet to some extent. Nevertheless, the present method reproduces the deformation of the droplet during the coalescence quite well (very similar to the experiment) and gives overall satisfactory predictions (no worse than the other method in [54]).

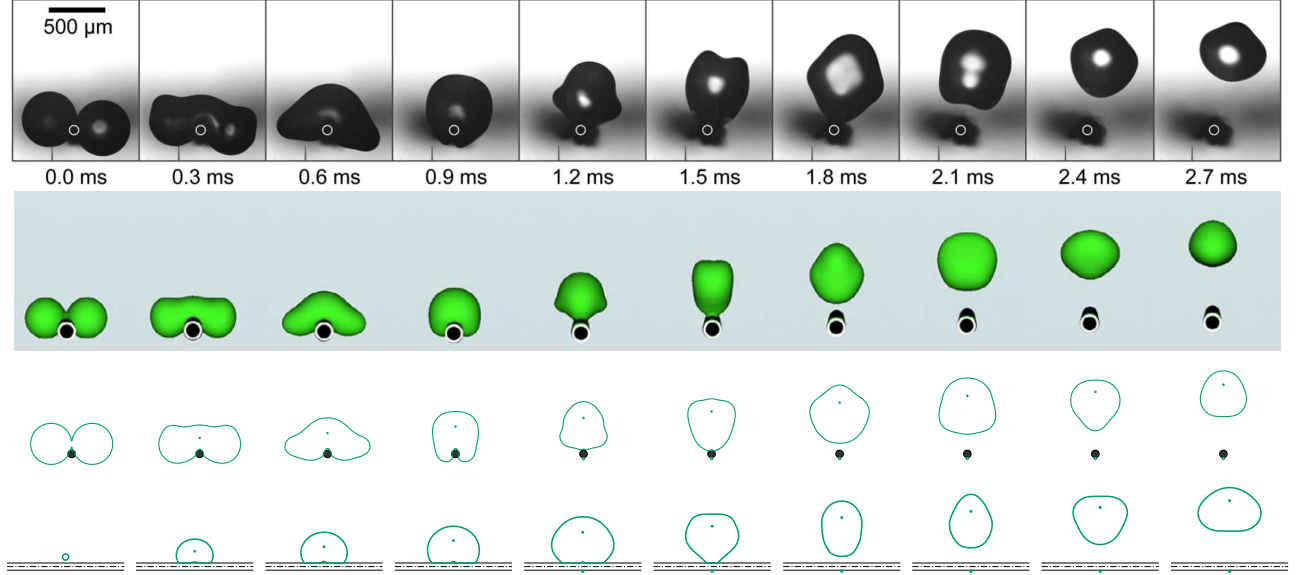


Figure 14: Evolution of the droplet shape for the 3D coalescence-induced droplet jumping from a hydrophobic fiber. The first row is from the experiment in [53] (Reproduced with permission from Phys. Rev. Lett. 115, 074502 (2015). Copyright 2015 American Physical Society). The second row is from the simulation in [54] (Reproduced with permission from Int. J. Multiphase Flow 101, 35 (2018). Copyright 2018 Elsevier Ltd.). The third and fourth rows are by the present simulations (the third row shows the interfaces in the $x - y$ plane at $z \approx 0$, and the fourth shows the interfaces in the $y - z$ plane at $x \approx 0$). The left half ($x < 0$) and back half ($z < 0$) of the present results are added by using the symmetric conditions. The times for the present ten snapshots from left to right are 0, $0.6517T_{ci}$, $2 \times 0.6517T_{ci}$, $\dots$, $9 \times 0.6517T_{ci}$, which correspond to the same moments taken in the experiments.

Figure 15 shows twenty snapshots of the droplets for the asymmetric cases by the present simulation and from the experiments of [53] at the same (dimensionless) times. One can see that the drop shape evolution is very similar to the corresponding photos taken in the experiments despite that a larger Oh is used here. This confirms that the simulated problem is still in the capillary-inertial regime.

4 Concluding Remarks

In summary, we have proposed a simplified method to apply WBC on curved walls for conservative PF LBM simulations of two-phase flows. Due to its special use of the PF profile across the interface, this method can avoid complex interpolations even for 3D cases. The incorporation of the LBM for the CACE extends its capability to multiphase problems with large density and viscosity ratios while maintaining the simplicity of the

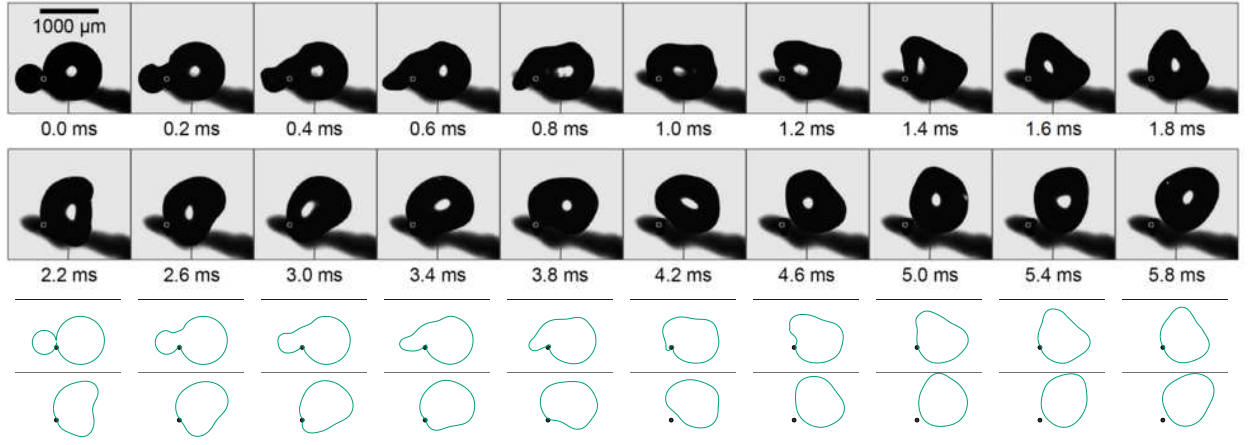


Figure 15: Evolution of the droplet shape for the droplet jumping from a hydrophobic fiber upon the coalescence of two droplets of radii R and $R_2 = 2R$. The first and second rows are from the experiment in [53] (fig. S7 therein). (Reproduced with permission from Phys. Rev. Lett. 115, 074502 (2015). Copyright 2015 American Physical Society). The third and fourth rows are the interfaces in the $x - y$ plane at $z \approx 0$ by the present simulations. To better match the experimental setup, the present results are rotated by 180° anticlockwise with respect to the line at $x = x_f, z = 0$. The times for the present ten snapshots in the third row are $0, 0.4217T_{ci}, 2 \times 0.4217T_{ci}, \dots, 9 \times 0.4217T_{ci}$, and those in the fourth row are $11 \times 0.4217T_{ci}, 13 \times 0.4217T_{ci}, 15 \times 0.4217T_{ci}, \dots, 29 \times 0.4217T_{ci}$ (from left to right).

overall formulation. The proposed method was used to study the 2D drop spreading on an embedded plate, the axisymmetric drop spreading on a sphere and on an oblate/prolate spheroid, the axisymmetric drop impact onto a sphere as well as the 3D coalescence-induced droplet jumping from a round fiber. The results by the present method agree well with numerical results using other WBCs, theoretical predictions, previous simulation results or corresponding experimental observations. At the same time, the method still needs further enhancement to overcome instability problems for drop impact at very high Reynolds numbers (very low Ohnesorge numbers). Besides, the use of the static contact angle may become insufficient to capture the drop dynamics for certain cases and more advanced dynamic contact angle model may be required. These are left for future work.

A Analytical solution for a drop on an oblate spheroid

In this part, the equations for calculating the analytical solution for a drop on an oblate spheroid with a contact angle is θ_w are derived (the situation on a prolate spheroid is similar and thus omitted). The initial radius of the drop R before touching the oblate spheroid is known. Figure 16 shows the equilibrium configuration in the cylindrical coordinate of a drop on an oblate spheroid with the relevant quantities illustrated. In the $z - r$ plane the surface of the oblate spheroid becomes an ellipse. When the origin O is put at the center of the ellipse, the equation for it may be written as,

$$\frac{z^2}{R_z^2} + \frac{r^2}{R_r^2} = 1, \quad (\text{A.1})$$

where $R_r > R_z$. In equilibrium the drop takes a spherical shape with its center at the point O_1 and its radius is R_d . Suppose the left contact line is at point P with the coordinate $(z_P, r_P) = (R_P \sin \alpha, R_P \cos \alpha)$. The angle formed by $\overline{O_1 P}$ and the r -axis is β . The green solid line in fig. 16 denotes the tangent line of the ellipse at P and the angle γ is formed by the tangent line and the r -axis. In total, there are five unknown quantities: R_P , α , β , γ and R_d . We seek to establish five independent equations for them. By substituting the coordinate of P into eq. (A.1), one has,

$$R_P^2 \left(\frac{\sin^2 \alpha}{R_z^2} + \frac{\cos^2 \alpha}{R_r^2} \right) - 1 = 0. \quad (\text{A.2})$$

Besides, as found in fig. 16, $r_P = R_P \cos \alpha = R_d \cos \beta$, thus one has,

$$R_P \cos \alpha - R_d \cos \beta = 0. \quad (\text{A.3})$$

From this figure, one can also obtain,

$$\theta_w + \gamma - \frac{\pi}{2} = \beta. \quad (\text{A.4})$$

Along the tangent line passing P one can find by using eq. (A.1) that,

$$\left. \frac{dr}{dz} \right|_P = - \left(\frac{R_r^2}{R_z^2} \right) \frac{z_P}{r_P} = - \left(\frac{R_r^2}{R_z^2} \right) \tan \alpha = - \frac{1}{\tan \gamma},$$

from which one gets,

$$\left(\frac{R_r^2}{R_z^2} \right) \tan \alpha \tan \gamma - 1 = 0. \quad (\text{A.5})$$

From the conservation of drop volume, one gets,

$$\frac{4}{3} \pi R^3 = V_{cap,s} - V_{cap,e}, \quad (\text{A.6})$$

where $V_{cap,s}$ and $V_{cap,e}$ are the volumes of the spherical cap (the part of the sphere with $z \geq z_P$) and the ellipsoid cap (the part of the oblate spheroid with $z \geq z_P$), which can be calculated as,

$$V_{cap,s} = \frac{\pi}{3} h_s^2 (3R_d - h_s), \quad V_{cap,e} = \frac{\pi}{3} \left(\frac{R_r^2}{R_z^2} \right) h_e^2 (3R_z - h_e),$$

with $h_s = R_d(1 + \sin \beta)$ and $h_e = R_z - R_P \sin \alpha$ being the heights of the spherical cap and the ellipsoid cap, respectively. With eqs. (A.2), (A.3), (A.4), (A.5) and (A.6), one can solve for the five unknowns by Newton's method (for brevity, the details are omitted).

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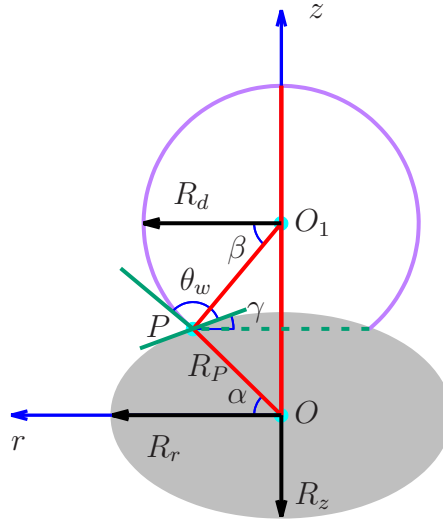


Figure 16: The equilibrium configuration for a drop on an oblate spheroid in the cylindrical coordinate.

was supported by NSFC (Grant No. 12102071).

Data availability

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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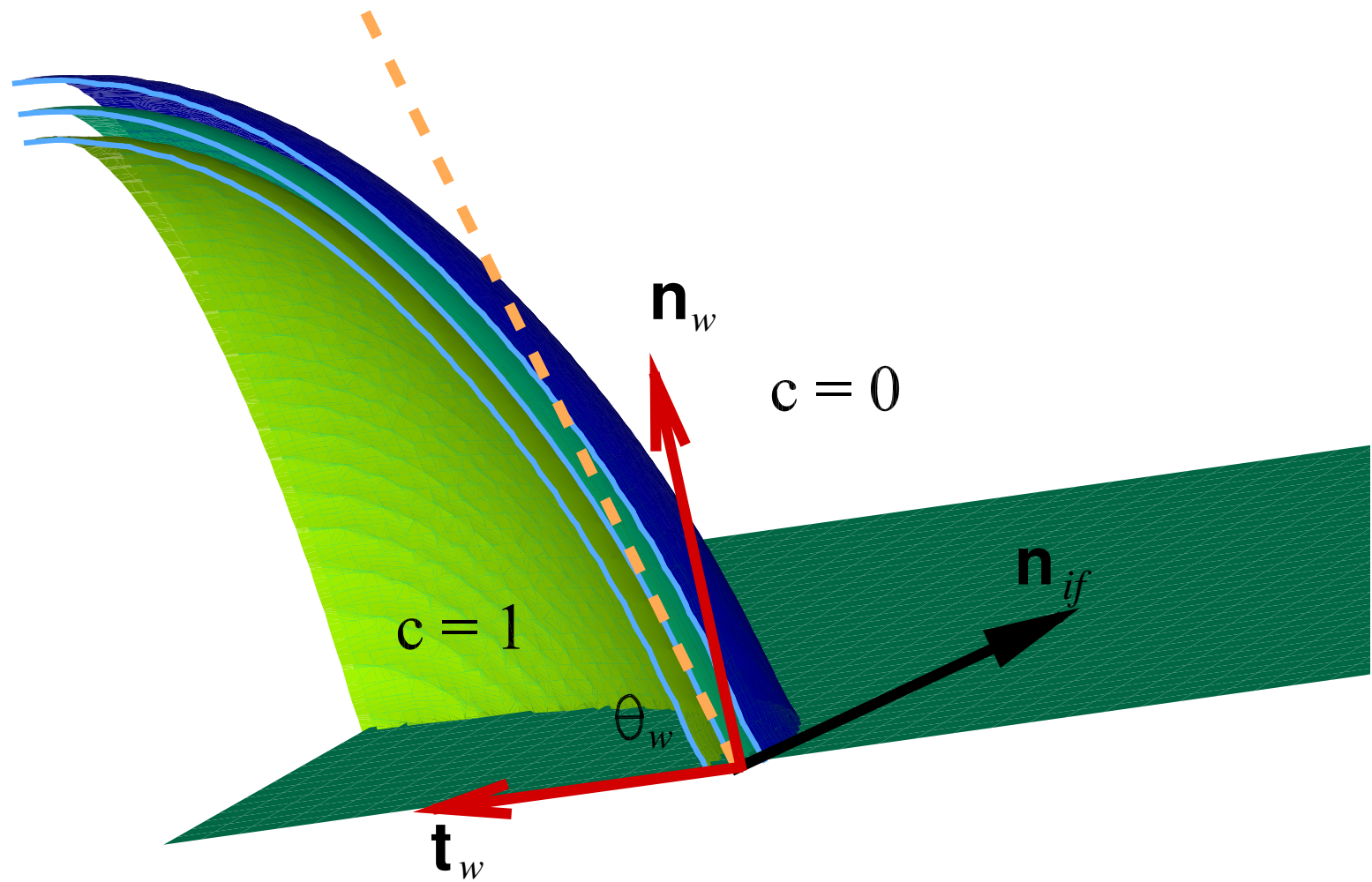
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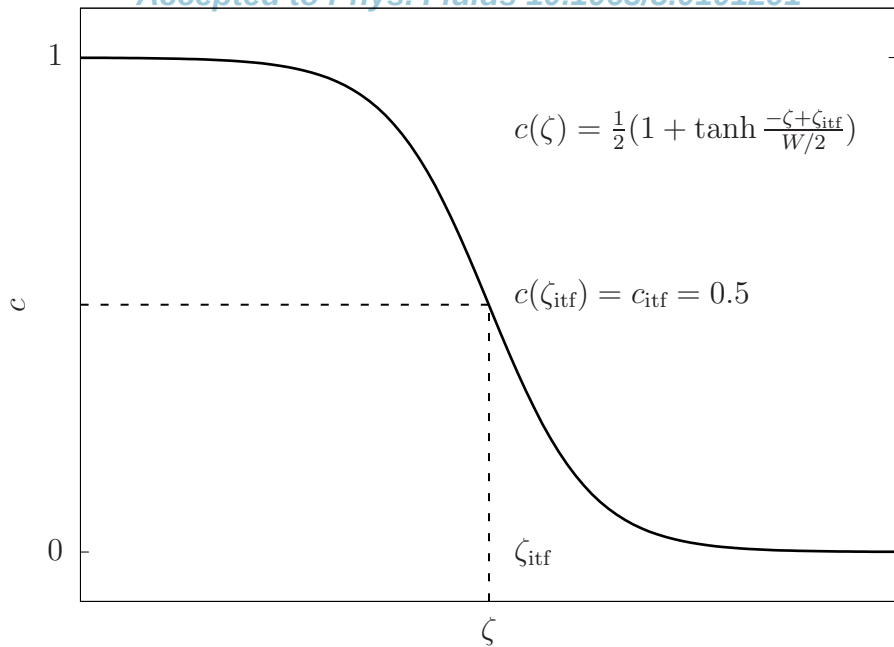
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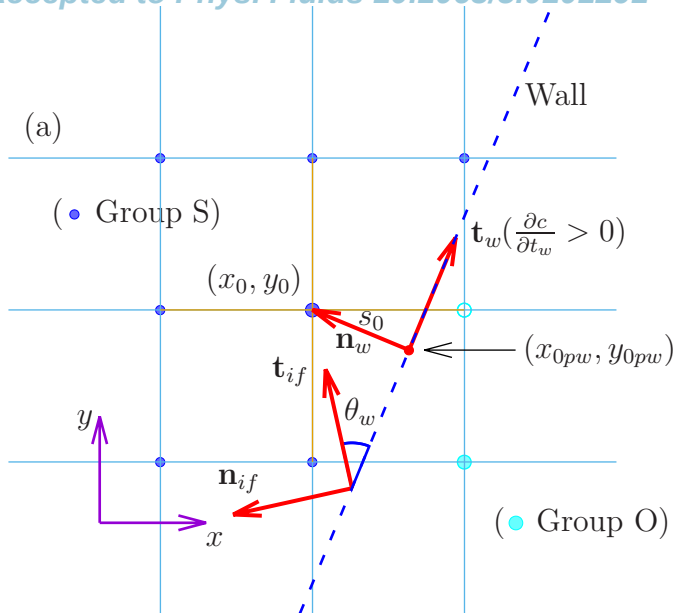
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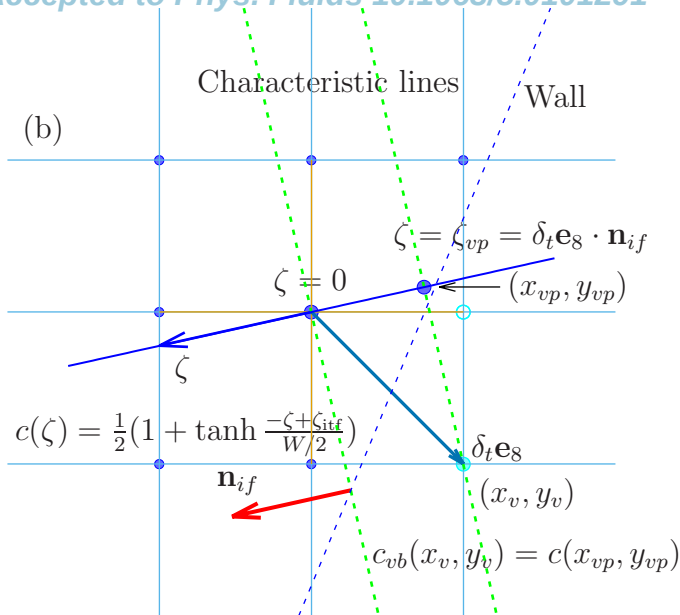
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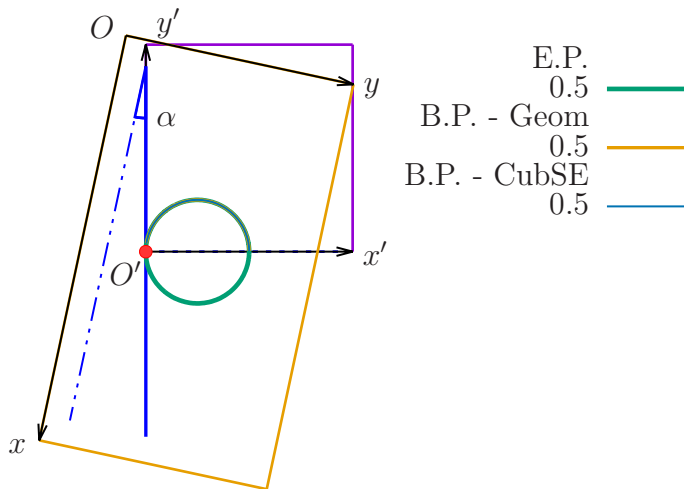
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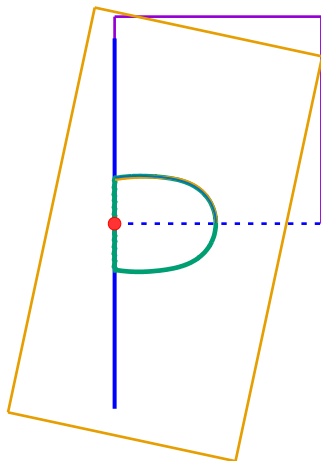


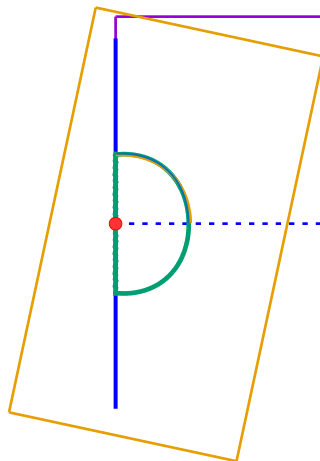


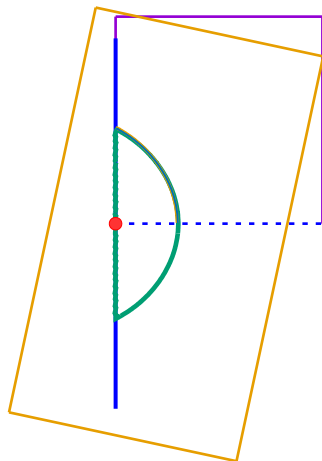


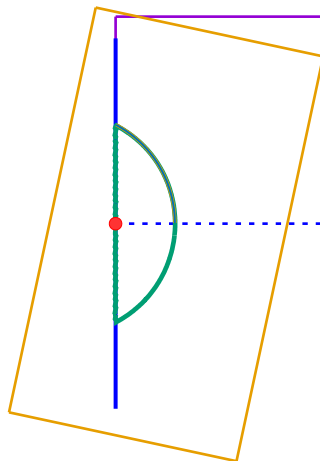


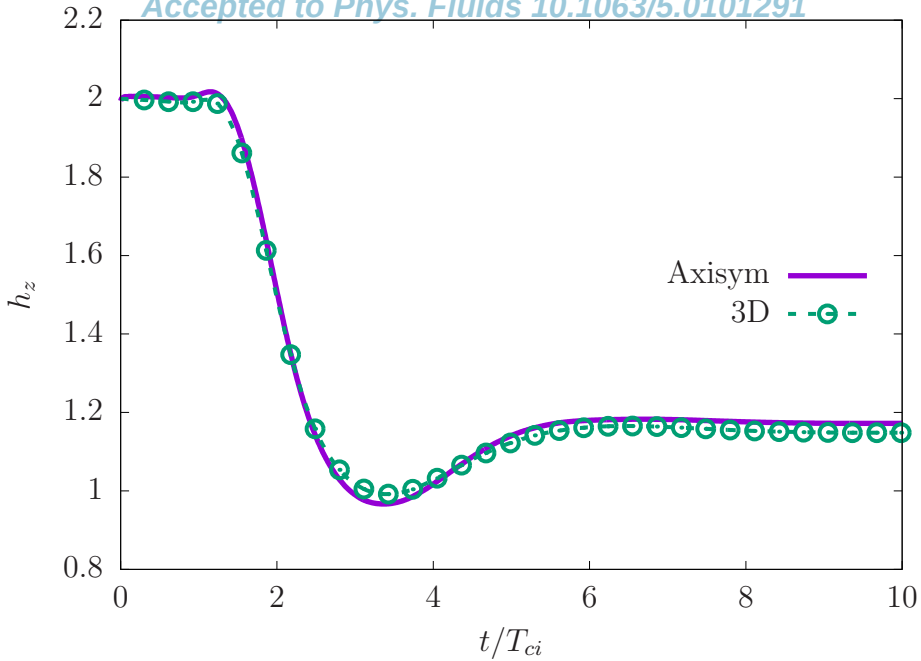


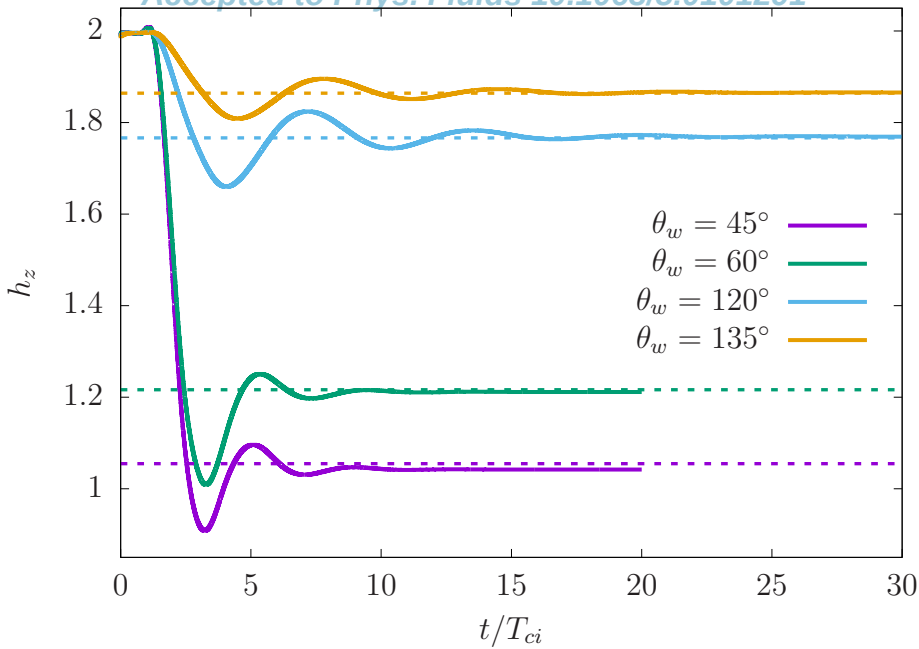


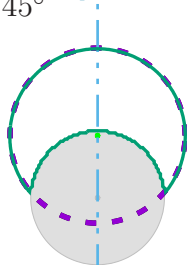
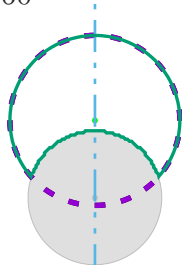
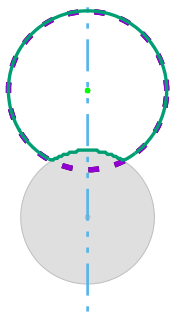
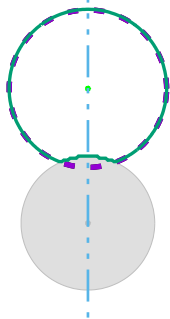


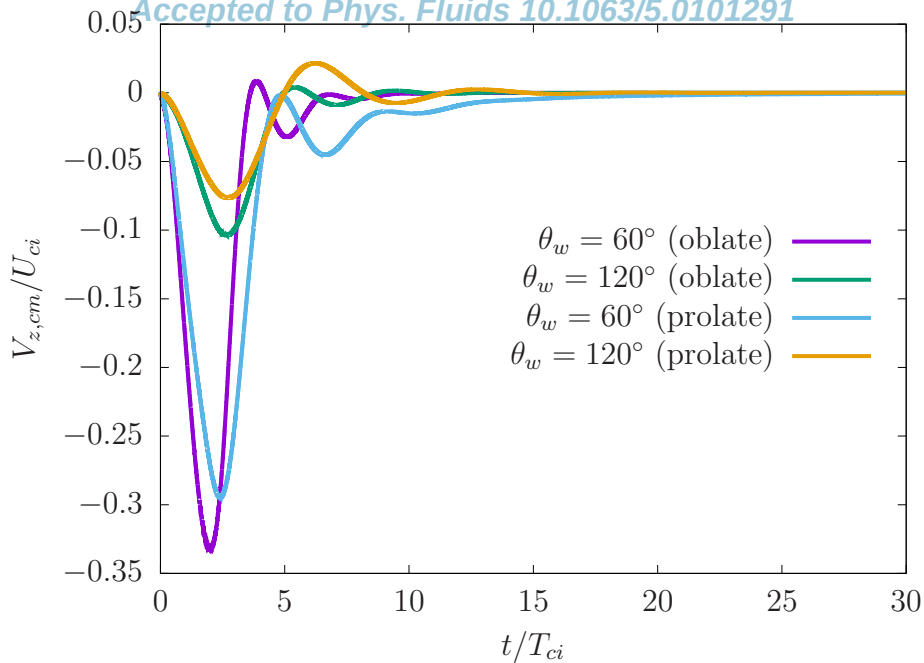


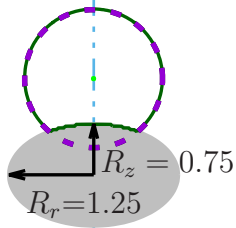
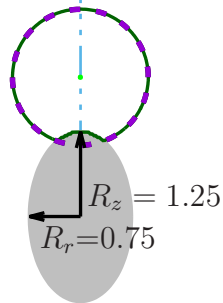
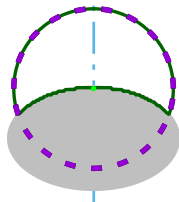
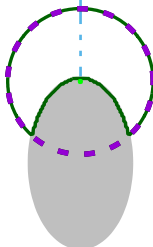


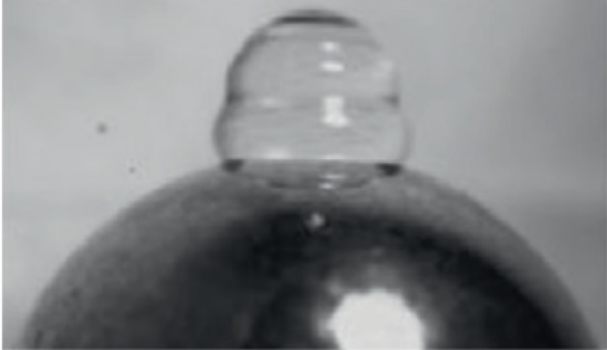




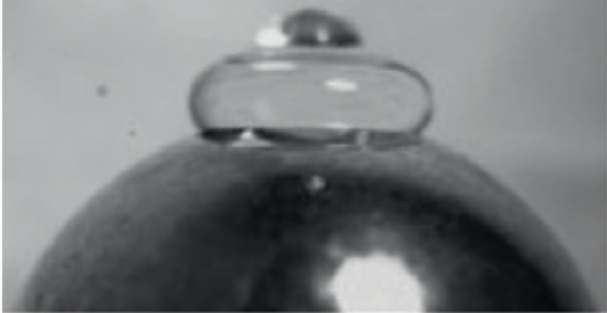
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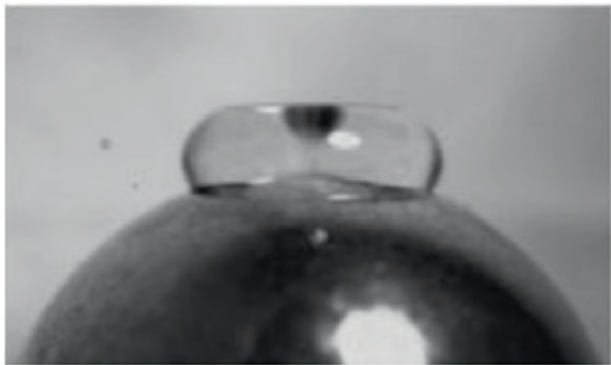


(a) 120° (b) 120° (c) 60° (d) 60° 



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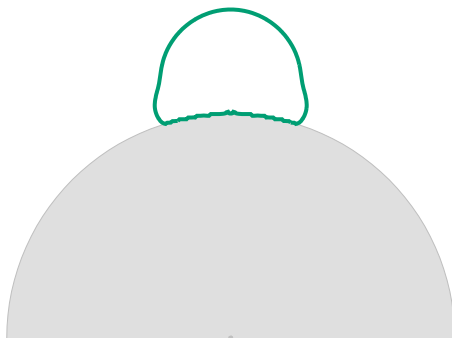




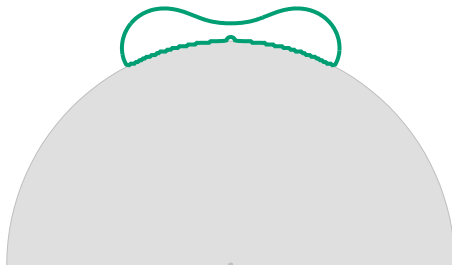


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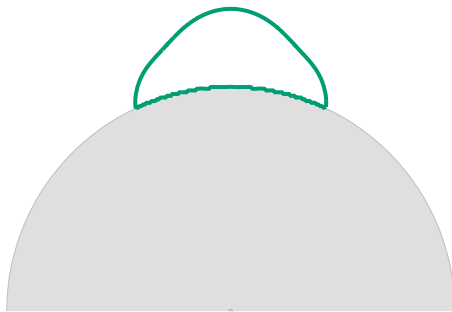


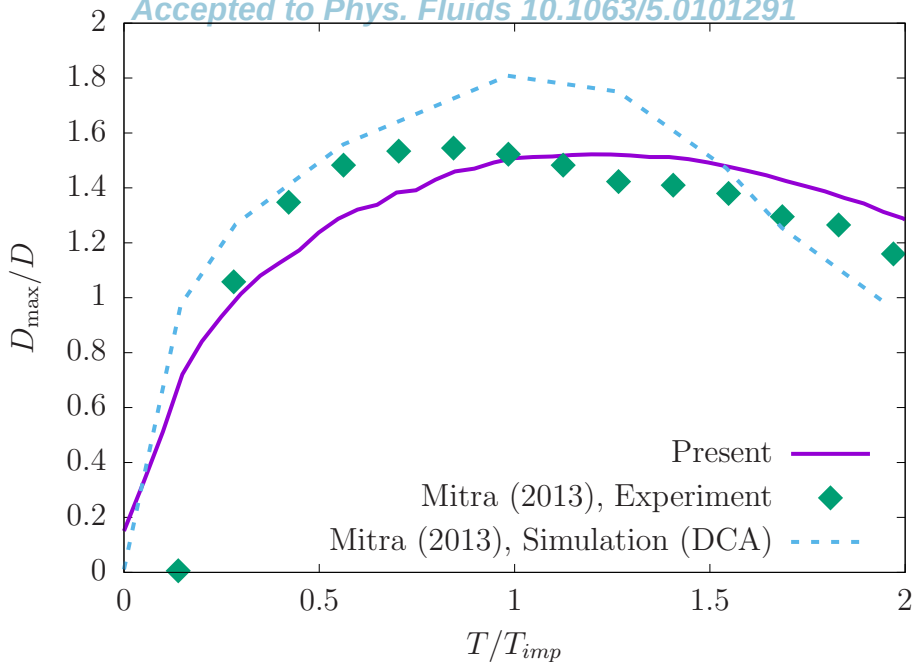




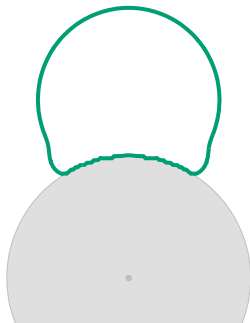


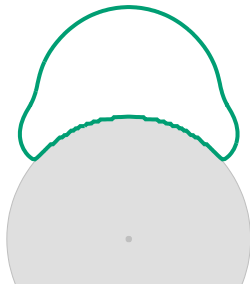


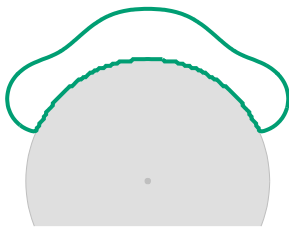


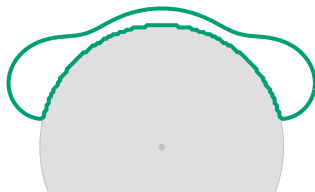


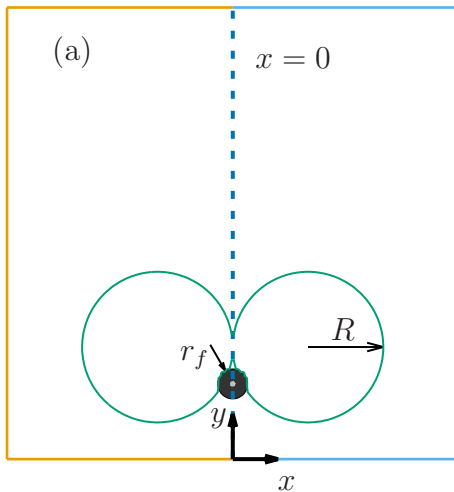


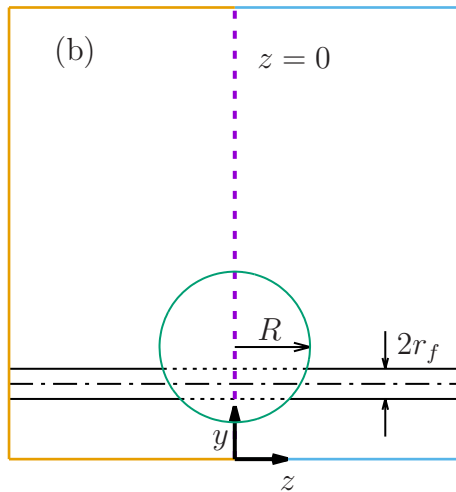


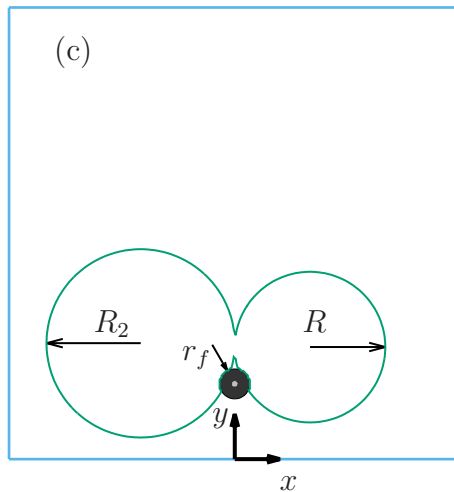


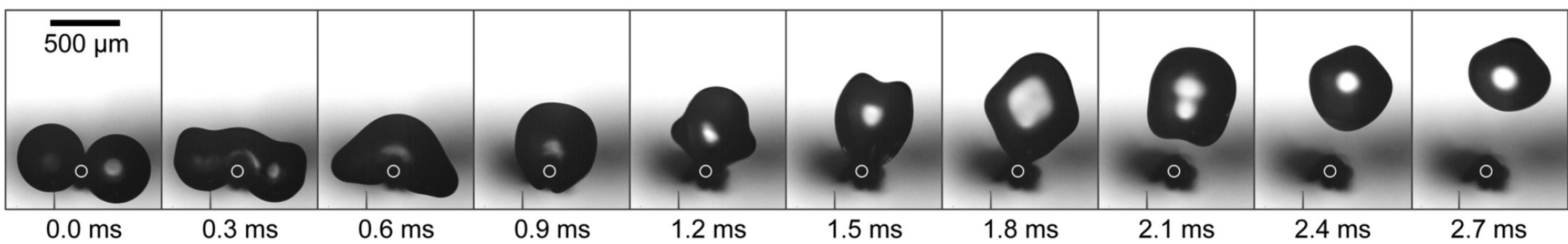


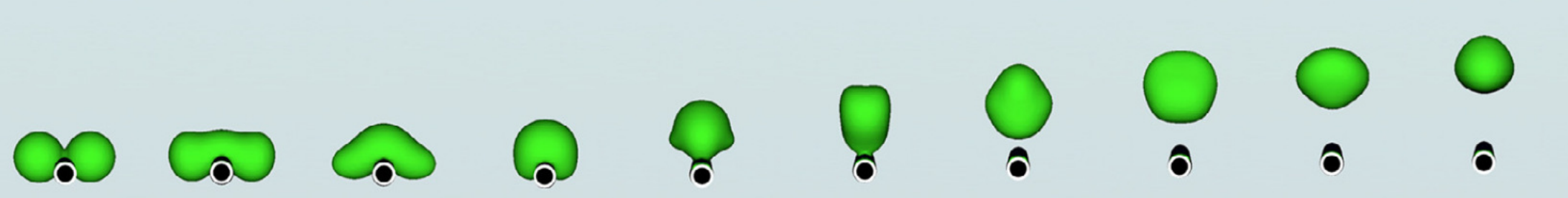


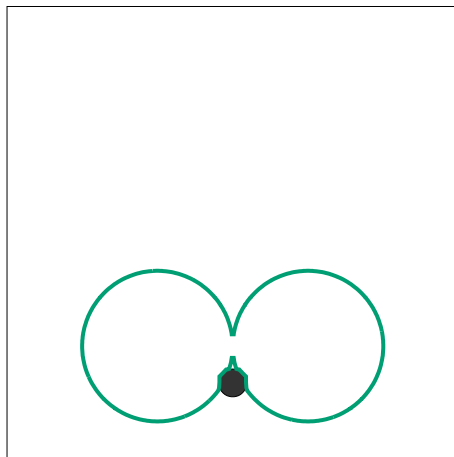


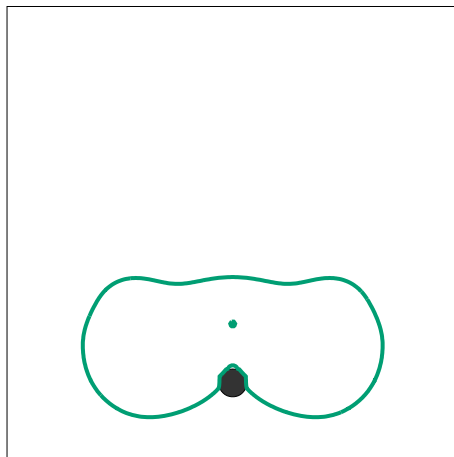


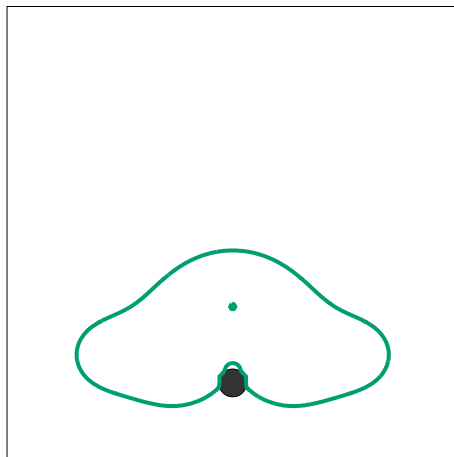


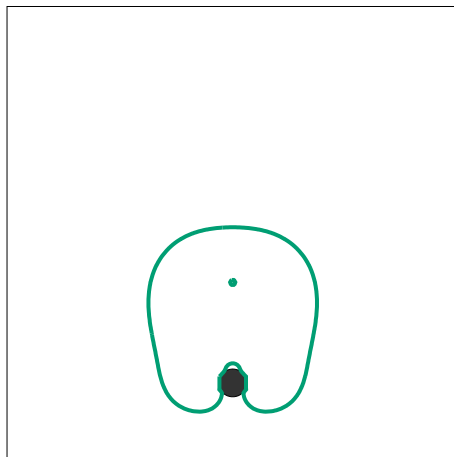


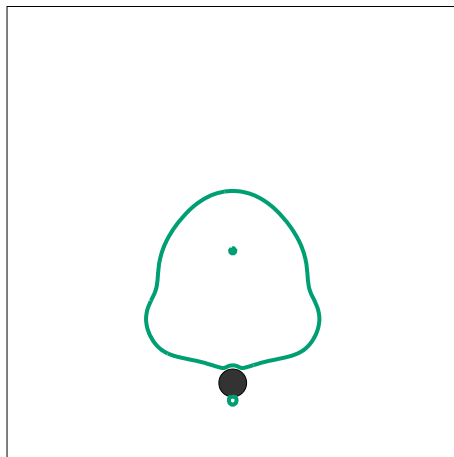


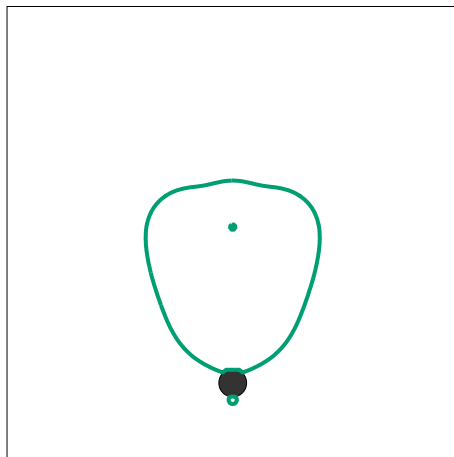


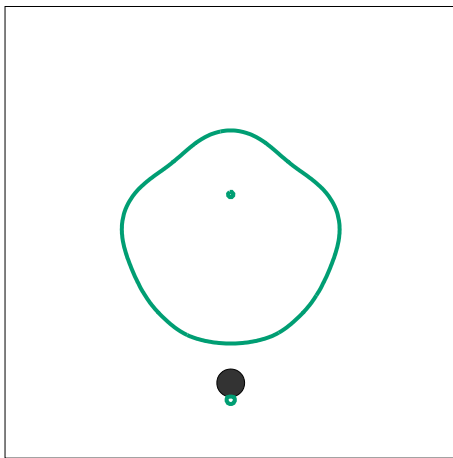


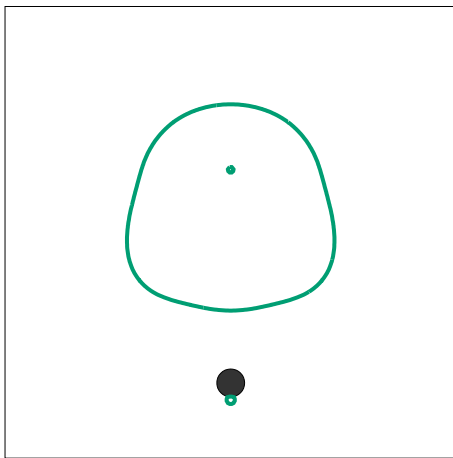


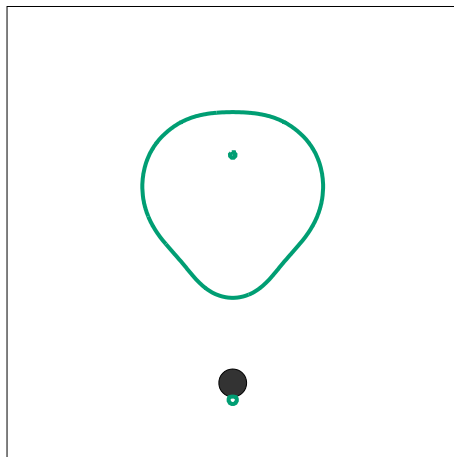


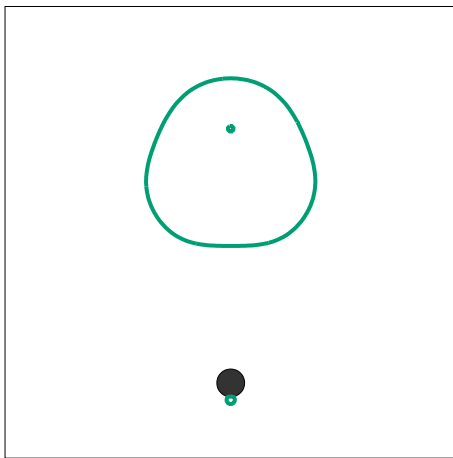




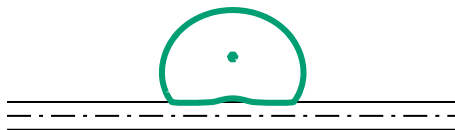


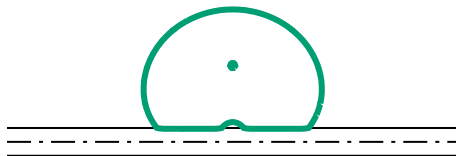


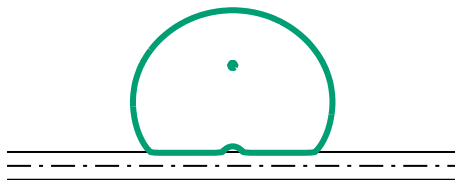


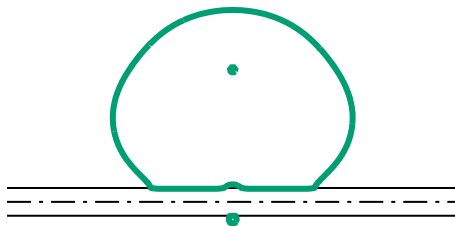


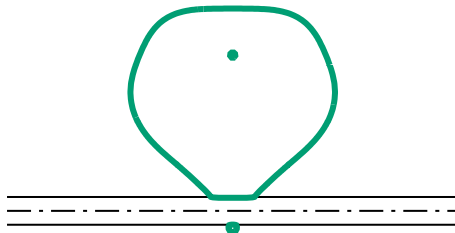


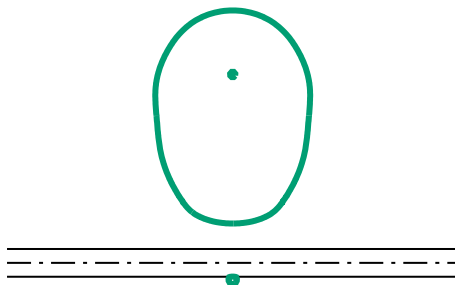


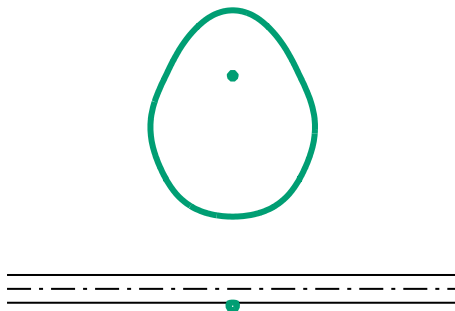


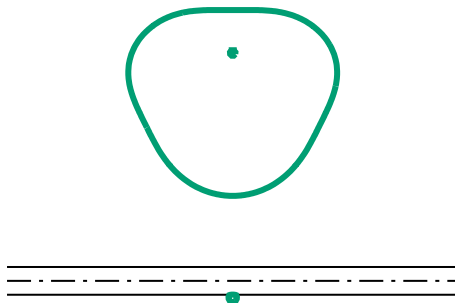


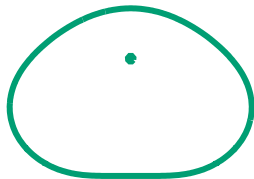


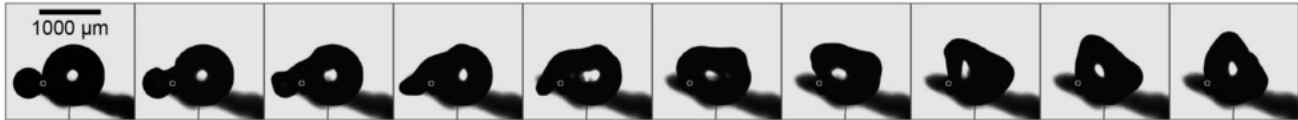












0.0 ms

0.2 ms

0.4 ms

0.6 ms

0.8 ms

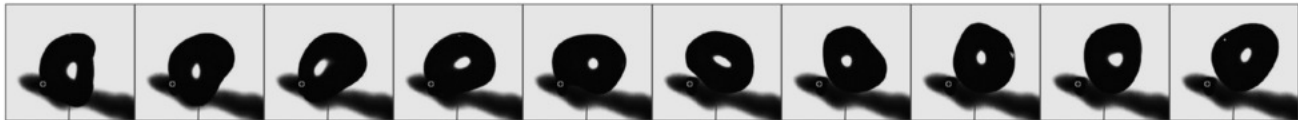
1.0 ms

1.2 ms

1.4 ms

1.6 ms

1.8 ms



2.2 ms

2.6 ms

3.0 ms

3.4 ms

3.8 ms

4.2 ms

4.6 ms

5.0 ms

5.4 ms

5.8 ms

